arXiv:2503.24377v3 [cs.CL] 8 Sep 2026

Harnessing the Reasoning Economy: A Survey of Efficient Reasoning for Large Language Models

RUI WANG*, The Chinese University of Hong Kong, Hong Kong SAR, China
HONGRU WANG*, University of Edinburgh, Edinburgh, UK
BOYANG XUE*, The Chinese University of Hong Kong, Hong Kong SAR, China
YIXIA LI†, Southern University of Science and Technology, China
JIANHUI PANG†, University of Macau, Macau SAR, China
SHUDONG LIU†, University of Macau, Macau SAR, China
YI CHEN, The University of Hong Kong, Hong Kong SAR, China
JIAHAO QIU, Princeton University, USA
DEREK FAI WONG, University of Macau, Macau SAR, China
GUANHUA CHEN, Southern University of Science and Technology, China
HENG JI, University of Illinois Urbana-Champaign, USA
KAM-FAI WONG, The Chinese University of Hong Kong, Hong Kong SAR, China

Recent advancements in Large Language Models (LLMs) have significantly enhanced their ability to perform complex reasoning tasks, transitioning from fast and intuitive thinking (System 1) to slow and deep reasoning (System 2). While System 2 reasoning improves task accuracy, it often incurs substantial computational costs due to its slow thinking nature and inefficient or unnecessary reasoning behaviors. In contrast, System 1 reasoning is computationally efficient but leads to suboptimal performance. Consequently, it is critical to balance the trade-off between performance (*benefits*) and computational costs (*budgets*), giving rise to the concept of *reasoning economy*. In this survey, we provide a comprehensive analysis of the reasoning economy by structuring the landscape into a lifecycle-based taxonomy: (1) Sources of Inefficiency: We first analyze the mechanisms of reasoning inefficiency, defining where the static allocation of computation fails to align with dynamic task complexity. (2) Post-training Optimization (Reasoning Capabilities Supply-side): We review methods for constructing high-density reasoning assets, focusing on building models with inherent efficiency before deployment. (3) Inference Optimization (Reasoning Capabilities Demand-side): We explore dynamic system designs for adaptive budget management, where computational resources are allocated on-demand to maximize the return on inference costs. We conclude that advancing current

*Project Lead and Equal Contributions.
†Significant Contributions.

Authors' Contact Information: Rui Wang, The Chinese University of Hong Kong, Hong Kong SAR, China, rwang@se.cuhk.edu.hk; Hongru Wang, University of Edinburgh, Edinburgh, UK, hrwang@se.cuhk.edu.hk; Boyang Xue, The Chinese University of Hong Kong, Hong Kong SAR, China, byxue@se.cuhk.edu.hk; Yixia Li, Southern University of Science and Technology, China; Jianhui Pang, University of Macau, Macau SAR, China, nlp2ct.pangjh3@gmail.com; Shudong Liu, University of Macau, Macau SAR, China, nlp2ct.shudong@gmail.com; Yi Chen, The University of Hong Kong, Hong Kong SAR, China; Jiahao Qiu, Princeton University, USA; Derek Fai Wong, University of Macau, Macau SAR, China; Guanhua Chen, Southern University of Science and Technology, China; Heng Ji, University of Illinois Urbana-Champaign, USA; Kam-Fai Wong, The Chinese University of Hong Kong, Hong Kong SAR, China.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2018 Copyright held by the owner/author(s). Publication rights licensed to ACM.
Manuscript submitted to ACM

LRMs requires a synergistic approach: effective post-training methods to enhance **intrinsic per-token intelligence**, coupled with adaptive test-time strategies to fully **unlock model potential**. This dual optimization is essential to approach performance ceilings while minimizing computational costs. By offering insights and highlighting open challenges, we aim to shed light on strategies for improving the reasoning economy of LLMs, thereby serving as a valuable resource for advancing research in this evolving area[1].

CCS Concepts: • **Computing methodologies** → **Natural language processing**; **Natural language generation**.

Additional Key Words and Phrases: Large Language Models, Reasoning Economy, Post-training, Test-time Methods

ACM Reference Format:

Rui Wang, Hongru Wang, Boyang Xue, Yixia Li, Jianhui Pang, Shudong Liu, Yi Chen, Jiahao Qiu, Derek Fai Wong, Guanhua Chen, Heng Ji, and Kam-Fai Wong. 2018. Harnessing the Reasoning Economy: A Survey of Efficient Reasoning for Large Language Models. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 30 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Large Language Models (LLMs) have demonstrated exceptional performance across various language understanding and generation tasks, particularly with the advent of Chain-of-Thought (CoT) prompting [188], which encourages models to generate explicit, step-by-step reasoning to arrive at the final answers. While LLMs excel in many scenarios, their reliance on fast, intuitive thinking often falls short when faced with complex reasoning challenges, such as advanced mathematics [6, 235] and coding tasks. Therefore, recent studies attempt to further boost the reasoning capabilities of LLMs, as exemplified by OpenAI o1 [126], DeepSeek R1 [33] and QwQ [145], to thoroughly explore potential solutions and improve problem-solving accuracy with slow and deep thinking [173], leading to the blossoming of Large Reasoning Models (LRMs) and new scaling law during the inference [161].

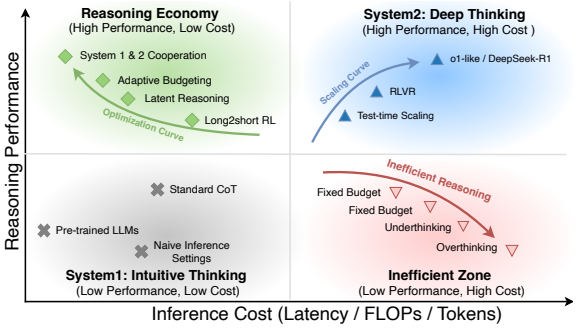


Fig. 1. The Landscape of Reasoning Economy

However, this cognitive upgrade comes with a prohibitive price tag. The deliberative nature of LRMs incurs substantial inference overhead due to the generation of elongated CoT sequences. A critical bottleneck in current deployments is the static, *one-size-fits-all* strategy: heavy reasoning is applied indiscriminately, regardless of task difficulty. This results in severe **resource misalignment**, where models waste computation on trivial queries (*overthinking*) [31, 192] while failing to allocate sufficient depth to truly intricate problems (*underthinking*) [161, 182]. This imbalance between capability and efficiency presents a significant challenge - achieving ***Reasoning Economy***, a global optimum that optimizes token usage (***budgets***) by emphasizing meaningful reasoning steps, reducing redundancy, and dynamically adjusting computational effort based on task complexity. Therefore, it not only reduces computation and latency of LRMs but also unlocks their potential by intelligently stopping or diving deeper (***benefits***) [3, 83]. With the growing importance of the reasoning economy, there is an urgent need to systematically understand and analyze different reasoning behaviors of LRMs, reveal potential challenges toward efficient LRMs, and clearly showcase solutions to achieve the reasoning economy.

[1] https://github.com/DevoAllen/Awesome-Reasoning-Economy-Papers

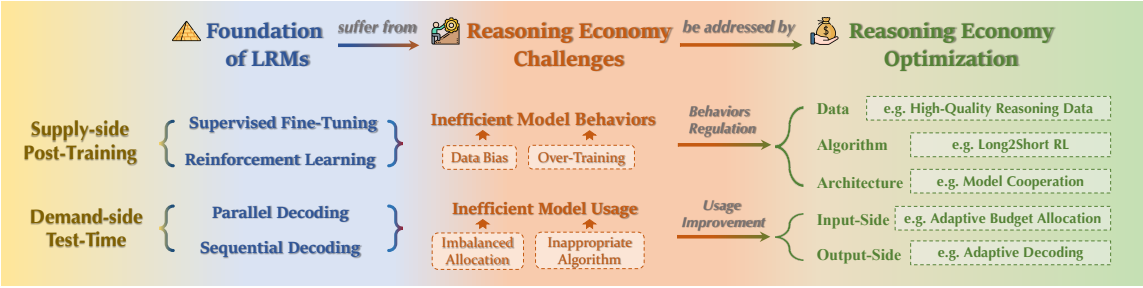


Fig. 2. We first provide a review of the Foundation of LRMs, including post-training and test-time methods. Then, we analyze the Challenges of the Reasoning Economy and survey the efforts to achieve the reasoning economy by regulating inefficient model behaviors and improving test-time budget allocation and utilization.

1.1 Review Structure

In this survey, we present the first comprehensive and systematic review of reasoning economy for LRMs. Specifically, we begin by establishing the *Foundation of LRMs* (§ 2), quickly dissecting how post-training methods (§ 2.1) (i.e, supervised fine-tuning and reinforcement learning) shape reasoning behaviors and how test-time strategies (§ 2.2) (i.e, parallel and sequential) influence the model performance. Building on this foundation, we conduct a rigorous analysis of *Challenges towards Reasoning Economy* (§ 3), classifying them into inefficient behaviors from the model itself (§ 3.1) or inefficient usage during test-time (§ 3.2). Finally, we discuss potential solutions for optimizing reasoning economy in terms of two directions: i) constructing high-efficiency assets by behaviors regulations in post-training(§ 4), which targets undesirable reasoning behaviors at their source in terms of data, algorithm and even model architecture, and ii) dynamic budget management systems in test-time (§ 5) to dynamically adapt computation. We also discuss several open challenges, and suggest future research directions (§ 6). By offering a clear and structured roadmap, our work aims to provide actionable insights to guide future research and foster the development of the reasoning economy for more sustainable LRMs.

1.2 Position and Contribution

Recent surveys have extensively covered the construction [98] and mechanisms [21] of LRMs. Concurrently, recognizing the computational costs associated with deep reasoning, several works [45, 143, 164, 222] have shifted focus toward *efficient reasoning*. Unlike system-level surveys focusing on pruning or distributed computing [115, 242], this line of work aligns with our focus on **reasoning efficiency**, optimizing the allocation of computational budgets during the reasoning process. However, despite this progress, current literature remains limited in three critical dimensions:

(1) Superficial Phenomenon Analysis: Prior works [143, 222] primarily concentrate on catalog symptoms of inefficiency, and methods to alleviate them, without dissecting their underlying *causes*, e.g.., data bias [47, 158] or reward hacking [160].

(2) Narrow Optimization Goals: Most surveys [45, 164] focus narrowly on *compression*, neglecting the broader paradigm of *dynamic allocation*, where efficiency implies investing *more* budget on hard tasks and *less* on easy ones.

(3) Fragmented Perspectives: Existing research often isolates training from inference [222]. There lacks a unified framework that views reasoning efficiency as a **lifecycle problem**, bridging the production of intelligence (*Supply*) with its adaptive consumption (*Demand*).

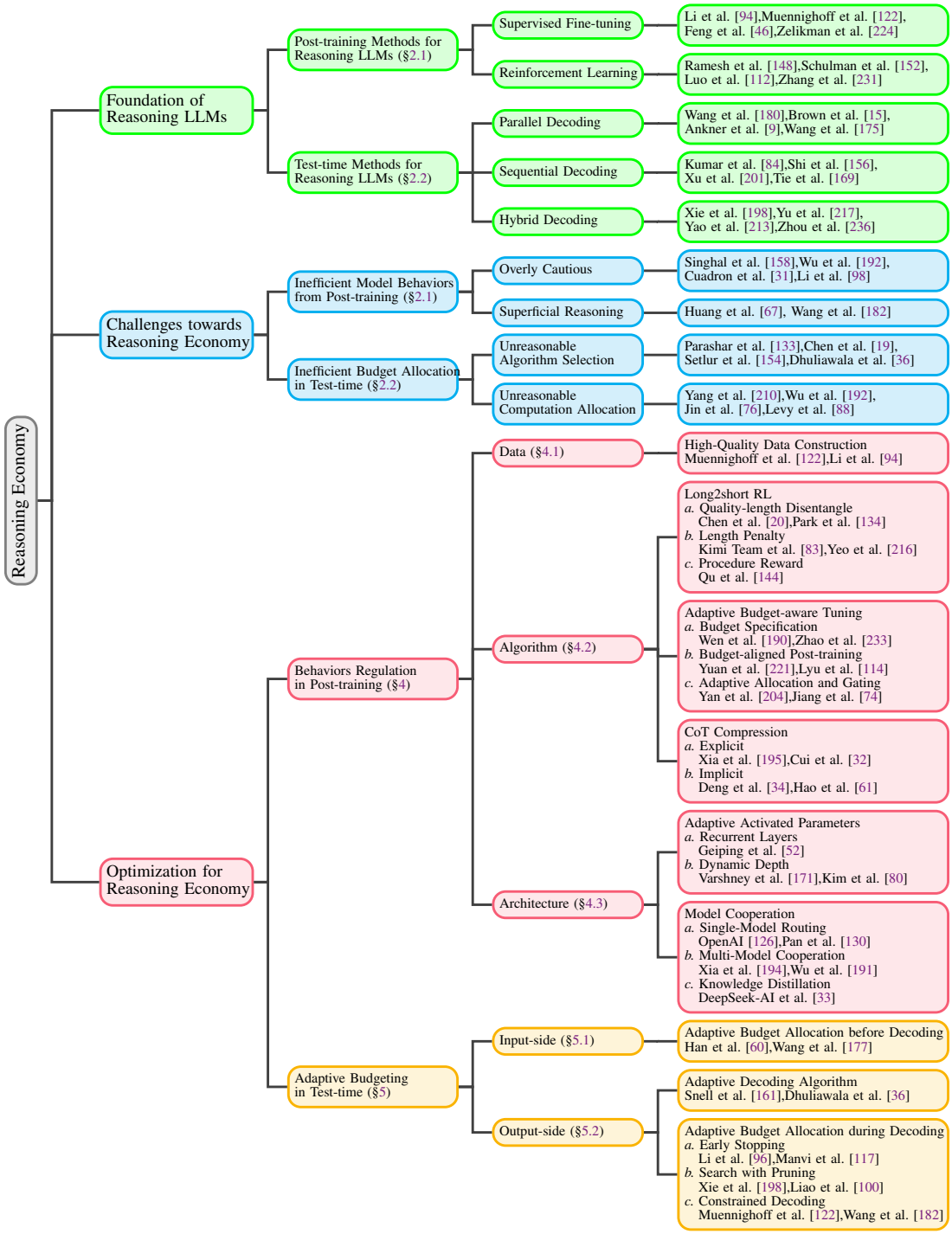


Fig. 3. Taxonomy of achieving Reasoning Economy, from foundation, challenges, to solutions.

2 Foundations of LRMs

This section outlines the fundamental methodologies that underpin the advanced reasoning capabilities of LRMs. We categorize these foundations into two stages: **Post-training** (§2.1), which shapes the intrinsic capabilities and behaviors of the model (the *supply* of intelligence); and **Test-time Scaling** (§2.2), which determines how these capabilities are utilized to approach performance ceilings through computational scaling (the *demand* management).

2.1 Post-training

Post-training is the critical phase where pre-trained base models are transformed into instruction-following reasoning agents. This process typically follows a two-stage pipeline: **Supervised Fine-Tuning (SFT)** to establish behavioral patterns [128, 187]. Building upon this foundation, **Reinforcement Learning (RL)** is then utilized to optimize the reasoning logic itself [13, 148, 152]. Crucially, RL enables the model to look beyond the training data, utilizing reward signals to explore diverse reasoning paths and self-correct errors, which is essential for aligning with rigorous logic and complex human preferences [33, 126].

Supervised Fine-tuning. SFT plays a crucial role in enhancing the zero-shot multi-task performance of LLMs [1] and serves as the foundational phase for initializing reasoning capabilities, a process often characterized as behavioral cloning. Consequently, the primary objective of the SFT stage is to internalize desired reasoning patterns by curating and incorporating target model behaviors into high-quality demonstration datasets. By leveraging high-quality task-specific data [168, 237], SFT refines the model's ability to generalize across various domains, improving both accuracy and reliability in real-world applications, such as summarization [39], machine translation [132], and question answering [85]. Recent studies have proposed leveraging self-improvement methodologies to enhance the reasoning capabilities of models [46, 57, 68, 79, 93, 163, 223, 224]. STaR [224] employs an iterative approach where the large LLM is prompted to generate multiple reasoning chains until the correct solution is obtained, after which the model is fine-tuned on the complete set of successful reasoning trajectories. The latest work Self-Reasoning Language Models (SRLM) [173] further extends to more general instruction-tuning datasets by mixing with few reasoning catalyst data, enabling SRLM to self-refine its own rationales iteratively. Though DeepSeek-AI et al. [33] proves that the SFT stage is no longer a must for LRMs by performing RL for pre-trained model, they also found that SFT contributes to accelerating the training process and achieving better performance for LRMs.

Reinforcement Learning. In the training of LRMs, Reinforcement Learning (RL) plays a key role in enhancing LLMs' reasoning capabilities by providing rewards for both the underlying reasoning process and the final answer [33, 126, 148, 152], rather than relying exclusively on token-by-token supervision. Moreover, recent studies [28] further indicate that RL-based approaches significantly enhance the generalizability of LLMs, allowing them to perform better in a wider range of tasks and scenarios. These findings underscore the importance of RL in refining model adaptability and improving reasoning capabilities. The core focus of reinforcement learning currently lies in the design of the reward signals. According to the different granularity of reward signals, there are two major reward models: the Process Reward Model (PRM) and the Outcome Reward Model (ORM). Specifically, the PRM [101] assigns rewards based on intermediate steps within an action sequence rather than solely on the final outcome [231]. PRM enables more fine-grained learning signals, guiding the LLMs toward optimal policies by rewarding beneficial intermediate behaviors. However, there are several limitations. One the one hand, the training data of PRM is challenging to obtain, as it either requires extensive

Category	Method	Extra Model	Signal	Decision Strategy
Parallel	Self-Consistency [180]	✗	None	Majority vote over sampled paths
	Shortest [37]	✗	Length	Select the shortest solution
	Best-of-N [9, 15]	✓	ORM	Select top-1 via outcome score
	Learned Aggregators [139]	✓	Learned	Merge candidates into new output
Sequential	Self-Refine [156, 225]	✗	Feedback	Generate → Self-feedback → Revise
	Self-Critique [84, 201]	✗	Critique	Iterative critique and revision
	Least-to-Most [209, 238]	✗	Prompt	Decompose into ordered sub-steps
Hybrid	Tree-of-Thought (ToT) [101, 213]	✓	PRM	BFS/DFS with state evaluation
	Guided Beam Search [175, 198]	✓	PRM	Beam expansion with pruning
	MCTS [106, 236]	✓	Value Func	Exploration via backpropagation

Table 1. Taxonomy of test-time scaling methods. We categorize methods into Parallel (independent sampling), Sequential (iterative refinement), and Hybrid (structured search) paradigms.

human annotations [101] or a large amount of sampling [175], which limits its application. On the other hand, they may also be overly stringent for the reasoning capabilities of LLMs [28].

In contrast, Outcome Reward Models (ORMs) [30] evaluate reasoning chains solely based on the correctness of the final result [9, 112, 199, 205]. This approach is foundational to *Reinforcement Learning with Verifiable Rewards* (RLVR), particularly in deterministic domains like mathematics and coding, where ground truth is easily verifiable. Consequently, ORMs facilitate scalable training via rule-based outcome reward functions (e.g., format and answer matching), as utilized in DeepSeek-AI et al. [33] and Luong et al. [113]. Although ORMs provide sparse supervision signals, they foster exceptional reasoning capabilities by allowing the model to freely explore the solution space without imposing rigid constraints on the reasoning process. A notable example is the DeepSeek-R1 model, which demonstrated emergent "Aha moments", spontaneous self-refinement and error correction, driven purely by rule-based accuracy rewards [33].

In summary, the choice between PRMs and ORMs represents a fundamental trade-off between supervision density and scalability. PRMs offer granular, step-by-step guidance that significantly narrows the search space, thereby facilitating faster and more stable policy convergence. In contrast, ORMs are implementation-friendly and provide the unconstrained exploration freedom necessary for models to discover non-trivial reasoning paths, such as emergent self-correction, that may deviate from human-annotated trajectories. Hence, developing hybrid reward frameworks that synergize the dense oversight of PRMs with the scalable exploration of ORMs remains a critical open challenge in the pursuit of more efficient and cost-effective LRMs.

2.2 Test-time Methods

Test-time methods aim to increase the compute for LLMs at test time to get more accurate and reliable results without post-training and often lift the LRMs to compare with further post-training [161]. We classify the test-time methods into Parallel, Sequential, and Hybrid methods, following Muennighoff et al. [122], Snell et al. [161].

Parallel Decoding. Parallel test-time methods operate by generating multiple candidate reasoning trajectories and employing a selector or aggregator to derive the final answer. Consequently, the design of the selection mechanism is central to these paradigms. The most straightforward approach is majority voting, popularized by Self-Consistency [180].

More sophisticated strategies utilize outcome-based scoring or re-ranking, such as Best-of- N selection and weighted voting, often guided by Outcome Reward Model (ORM) signals [9, 15, 175, 199, 205]. Beyond heuristic rules, recent research explores learned aggregators that directly synthesize a final prediction from the sampled set [139], or frames parallel scaling as population optimization through iterative evolutionary dynamics [229]. Notably, even lightweight heuristics, such as selecting the shortest solution, have demonstrated competitive performance [37]. Ultimately, these methods leverage the diversity of candidate solutions to enhance both reasoning accuracy and robustness.

Sequential Decoding. Sequential test-time methods allocate test-time compute along a single evolving trajectory. A representative pattern is Self-Refine[116], which alternates generation, self-feedback, and revision. Recent work further strengthens this process through training or step-level verification[156, 225]. Reflexion[157] extends self-refinement from single outputs to long-horizon agent behavior by storing natural-language reflection summaries of failed trials as episodic memory, which is then reused in subsequent attempts as a lightweight form of verbal reinforcement learning. To reduce ungrounded internal states, ReflAct further enforces explicit goal-state reflection by maintaining a structured belief state and assessing goal alignment before acting [81]. Beyond post-hoc correction, Least-to-Most prompting improves the trajectory by decomposing a hard problem into an ordered sequence of easier subquestions, optionally reusing cached reasoning templates [209, 238]. Nevertheless, evidence from CorrectBench suggests that self-correction gains are task- and difficulty-dependent and can introduce unnecessary latency or even hurt simpler cases [169], while the dynamics also differ between open-ended generation and multiple-choice settings [147].

Hybrid Methods. Hybrid methods combine sequential exploration with parallel proposal generation, using value or verification signals to guide branching and pruning. Guided beam search incorporates self-evaluation or outcome-supervised value models to rank partial candidates during decoding, improving planning in mathematical reasoning [198, 217]. Tree-structured search explicitly expands and evaluates intermediate "thought" nodes to enable lookahead and backtracking [213], while related tree-oriented aggregation such as ToM uses hierarchical composition to preserve coherence over long contexts [58]. More broadly, search can be paired with diversity-enhancing generation to populate candidate branches [150] and with value-guided MCTS-style exploration [106, 236]. In practice, these methods often rely on process reward models to score intermediate steps and select the most promising branches [101, 175, 232], thereby yielding high-quality reasoning trajectories.

Summary. Previous work found that the potential of LLMs is not fully reached, and the test-time methods are effective techniques to **approach the upper bound of LLMs**. Brown et al. [15] found that LLaMA-3-8B-Instruct can achieve 98.44% accuracy with 10,000 times repeated sampling and self-consistency, while only 82.9% with 100 samples. The state-of-the-art LRMs, e.g., o1 and R1, all exhibit inherent test-time scaling abilities, such as self-refinement, back-tracing, and thought-switching behaviors [216] in their extensive intermediate steps. Moreover, repeated sampling could further improve the performances of R1 [33] and R1-distilled LLMs. In particular, DeepSeek-R1-Distill-Qwen-14B [205] achieves 80% of accuracy on AIME24 [6] by applying majority voting on 64 samples, with only 69.7% accuracy of pass@1. Moreover, Snell et al. [161] found that the test-time methods are even more effective compared with additional training in easy and medium problems, while this is not the case on difficult problems.

3 Challenges Towards Reasoning Economy

In this section, we primarily examine the challenges in LRMs that impair the reasoning economy. We first analyze the Inefficient Model Behaviors caused by post-training methods, including Overly Cautious and Superficial Reasoning,

Category	Specific Behavior	Mechanism Analysis
Overly Cautious	**Length Bias** [20, 134, 158]	**Cause:** Reward models inherently prefer longer responses. **Symptom:** Outputs are verbose with low information density (low entropy).
	Overthinking [76, 107, 192]	**Cause:** Sparse rewards discourage risk-taking. **Symptom:** Excessive self-verification steps on simple tasks.
Superficial Reasoning	**Fake Thinking** [67, 182]	**Cause:** Underlying biases in SFT data or reward hacking. **Symptom:** Shallow imitation of reasoning structures (e.g., "Let's think") without logical validity.
	Underthinking [22, 182]	**Cause:** Inadequate exploration incentives. **Symptom:** Giving up too early or failing to explore complex branches on hard cases.

Table 2. A compact taxonomy of inefficient reasoning behaviors impacting the reasoning economy. We categorize these behaviors into structural redundancy (Overly Cautious) and qualitative failure (Superficial Reasoning).

which not only impair the LRMs' performance but also lead to significant computation waste. Then, we reveal the causes of waste of computation during test-time, the Inefficient Budget Allocation, to exhibit the importance of adaptive test-time setting to reduce computation while achieving higher performance.

3.1 Inefficient Model Behaviors from Post-training

LRMs are post-trained to align with human preferences and improve reasoning abilities. RL methods play a key role in the post-training stage for LRMs. However, RL optimization relies on reward models (RMs) that are inherently imperfect, primarily due to unreliable human preference annotations [47, 158]. Consequently, over-optimizing based on these flawed RMs can negatively impact the overall capabilities of LLMs. This situation exacerbates the risk of reward hacking [160], where models exploit loopholes in the reward function to maximize scores without genuinely aligning with the underlying human preferences. As a result, LLMs exhibit over-optimization [47], appearing to meet performance expectations during evaluation, but lacking a true understanding. Next, we will analyze previous research and summarize two notable behaviors that directly impact the reasoning efficiency of LLMs: Overly Cautious and Superficial Reasoning.

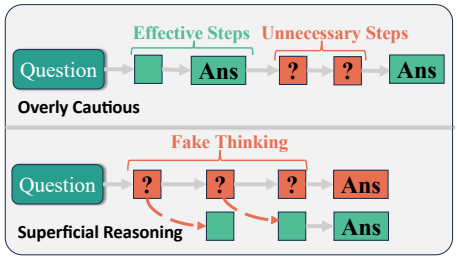


Fig. 4. Illustration of **Overly Cautious** and **Superficial Reasoning** Behaviors of LRMs. Overly cautious LLMs may repeatedly verify even after finding the correct answer. Fake thinking LLMs may show behaviors like self-refinement or backtracing. Yet, these behaviors are often superficial, not driven by a genuine awareness of needing to improve the answer.

One of the most prominent issues is **Overly Cautious**, where LLMs tend to generate longer responses that contain much redundant content, due to their hacking to maximize their reward scores. Previous research has shown that LLMs trained with RL tend to produce longer responses [162, 193] compared to those trained through SFT. As a result, several studies have aimed to answer two key research questions: *RQ1: What are the reasons for longer responses?* and *RQ2: Does the increased length indicate a bias or an enhancement of model capabilities?* Singhal et al. [158] discovered that in existing reward model training datasets, longer responses are often preferred (e.g., RLCD [207]: 63.1%), which leads to a length preference

in the RM (*RQ1*). Hence, the length-biased RM leads the LLMs to generate redundant content with little performance benefits, e.g., too many paraphrased or connection words. Furthermore, they found that using length as a proxy for reward models can yield performance comparable to that of PPO with RMs (*RQ2*), which further reveals the underlying bias and questions on the RM accuracy. Other studies have also indicated that RMs struggle to effectively disentangle length bias from response quality during RL training [20, 110, 134, 158].

3.1.1 Overly Cautious Behaviors of LRMs. DeepSeek-AI et al. [33] found that an accuracy- and format-based reward function of the RL process for LRMs is effective without the assistance of PRM or RM, and can largely avoid reward hacking. Thus, the Deepseek R1 exhibits a surprising improvement in complex reasoning tasks. Experiments [33, 83] suggest that such improvements from that the LRMs experience an "Aha" moment during the RL process: the reasoning path becomes longer and more complex, leading to behaviors such as self-refinement, recognizing and correcting mistakes, breaking down difficult steps, and iterating on alternative approaches(*R1*, *R2*). Though progress has been made, recent studies have also identified an "overly cautious" phenomenon [31, 76, 98, 192] in LRMs, e.g., with the naming as overthinking, characterized by excessive verification and redundant reasoning after giving the right answer (*R2*). This behavior stems from the assumption that longer outputs are more likely to contain the correct answer or appear more comprehensive, even when shorter, more concise responses would suffice. This overly cautious behavior not only results in inefficient token usage but also hampers LLM performance due to cumulated errors and the "lost in the middle" [89, 107].

Overly Cautious LRMs

LRMs exhibit excessive unnecessary verification and redundant reasoning on easy-to-handle questions or meaningless paraphrases and deviations, leading to inefficient token usage and increased computational costs.

3.1.2 Superficial Reasoning. Superficial Reasoning behavior refers to instances where LLMs appear to align with human preferences, but these behaviors either fail to produce tangible outcomes [183] or conceal other underlying objectives [56, 200]. For example, Greenblatt et al. [56] has demonstrated that LLMs may display differential behaviors across various demographic groups, which could only be found by comparing LLMs' responses to different people. Deceptive behavior is more challenging to detect than length bias or overly cautious behaviors, thus extensive and meticulous human observations [56, 182] are needed to identify it.

Fake Thinking LRMs. In the context of LRMs, recent studies have uncovered a **Fake thinking** behavior: They tend to generate plausible and fluent reasoning steps, but these trajectories are often unfaithful or lack logical rigor [155]. Some studies [67, 78, 182] found that LRMs may appear to engage in self-refinement or deliberate analysis, yet the generated chains can be post-hoc rationalizations or spurious reasoning that is difficult to detect from surface form alone [155]. Empirical evidence suggests that such behaviors often lead to little substantial progress: models may oscillate between alternatives, rewrite reasoning without convergence, or abandon high-quality steps early and replace them with ineffective refinements [182]. These deceptive behaviors not only waste test-time compute but can even hurt performance [54, 67]. Moreover, Anderson et al. [8] found that R1 can be poorly calibrated, giving up quickly on some solvable problems while getting stuck "thinking forever" on others. According to the reasoning boundary theory [22], fake thinking can also be treated as the meaningless attempts of LLMs while solving problems beyond their capabilities.

Superficial Reasoning LRMs

LRMs appear to work towards problem-solving, but their reasoning trajectories can be unfaithful or ineffective, leading to wasted computation and sometimes worse answers.

3.2 Inefficient Model Usage in Test-time

Though the test-time methods could push the performance of LRMs further in a training-free way, the application of test-time methods is often suboptimal [88, 96, 161]. Previous work found that two factors significantly influence test-time performance: (1) the selection of inference strategy (e.g., sampling, search, verification) [161], and (2) the test-time computation allocated to each question [88, 96].

3.2.1 Unreasonable Algorithm Selection. To employ a test-time method, there are two dimensions to be decided if we do not need to consider computation limitation: (1) which inference strategy to use and (2) what hyper-parameters to set (e.g., temperature and top-p). However, naive choices can waste computation and even degrade accuracy. In particular, a one-size-fits-all test-time inference strategy is rarely effective; the optimal choice depends on the task, model family, and problem difficulty (and often the compute budget) [2, 133].

For the choice of inference algorithm, Parashar et al. [133] found that there is not a single inference algorithm that suits all of the tasks. To be more specific, Chen et al. [19] found that majority voting in LLMs improves accuracy on simple problems but degrades performance on complex ones as votes increase. Similarly, Snell et al. [161] concluded that search-based methods outperform parallel methods on harder problems. Setlur et al. [154] argue that verifiers play a central role in effective test-time scaling. However, recent evidence suggests that under many practical inference budgets, scaling solution generation (e.g., Self-Consistency) can be more compute-optimal than scaling verification [159]. As for the parameter setting of the inference algorithm, recent work [36] found that a lower temperature of sampling is more suitable for reasoning tasks, while creative tasks need a higher temperature. Moreover, reasoning capabilities can be highly sensitive to subtle implementation choices such as decoding parameters, random seeds, and prompt formatting [64]. The above studies emphasize **the need for algorithmic adaptability based on task complexity**.

3.2.2 Unreasonable Computation Allocation. As mentioned before, though scaling the computation could bring consistent performance benefits, scaling LLaMA-3-8B-Instruct from generating 100 samples to 10,000 samples is often unacceptable for a simple question. However, a more complex problem is worth a high computation budget, while a small computation budget could lead to suboptimal accuracy. Recent studies propose the Reasoning Boundary [22] of LRMs, in which they found that middle-complexity problems need more computation. For sequential inference, several studies found that longer solutions with more self-refinement are not necessarily better [24, 76, 88, 192]; instead, there is an optimal refinement length that increases with problem difficulty [192]. Snell et al. [161] further found that compute-optimal allocation varies with problem complexity, and Yang et al. [210] reported that optimal length distributions can differ across domains. Moreover, reasoning models can be poorly calibrated on easy questions, generating excessive tokens with little accuracy gain, and decoding-time interventions can mitigate such overthinking [137]. Hence, inadequate computation can lead to wrong answers, while excessive computation may waste tokens and even impair performance. Above all, previous work empirically found that there is not a one-for-all computation setting for all tasks and samples varied in complexity. They emphasize **the importance of adaptive computation allocation based on task complexity**.

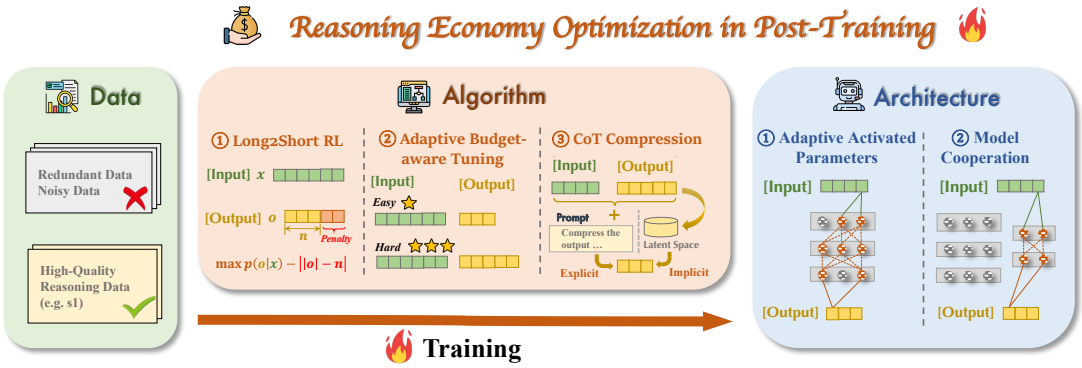


Fig. 5. Post-training Methods Optimization for Reasoning Economy.

4 Optimization for Reasoning Economy

part-1: Post-training

In this section, we provide an overview of post-training optimization methods aimed at enhancing reasoning economy in LLMs. These methods focus on regulating LLM behaviors to improve efficiency and reduce unnecessary computational overhead from Data, Algorithm, and Architecture perspectives.

4.1 Data

High Quality Data Construction. By explicitly encoding desired reasoning patterns and behaviors, high-quality post-training data can guide LLMs toward more advanced and effective reasoning [94, 122, 141, 237]. Qin et al. [141] utilized small-scale long-thought datasets sampled via test-time scaling [106, 236] and found that the resulting models exhibit clear long-thought reasoning patterns. Similarly, Muennighoff et al. [122] demonstrated that only 1,000 high-quality and diverse SFT samples can produce competitive LRMs comparable to advanced models such as o1-preview, highlighting the importance of data quality, diversity, and difficulty. Recent studies further suggest that data quality is also shaped by selecting a small set of high-impact examples that serve as effective cognitive templates for eliciting reasoning [94, 184, 215]. For example, Ye et al. [215] showed that strong reasoning can emerge from only a few carefully designed demonstrations, while Li et al. [94] found that a strategically selected subset of RL data can outperform much larger training sets. Moreover, Wang et al. [184] reported that even one-shot RLVR can yield substantial gains, highlighting the importance of sample selection and data efficiency in post-training.

4.2 Algorithm

Recent research has focused on three key strategies: optimizing reward structures to balance response length and quality (Long2short RL), enforcing explicit budget control during post-training and inference (Adaptive Budget-aware Tuning), and developing compression techniques to eliminate redundant reasoning steps while preserving accuracy (CoT Compression). These approaches collectively aim to enhance inference efficiency by either directly guiding models to generate concise responses or restructuring reasoning processes for more compact representations.

4.2.1 Long2short RL. To address the inefficiencies caused by length bias in RL-tuned LLMs, researchers have proposed various reward design improvements. For example, Singhal et al. [158] explored intuitive approaches, such

as increasing the KL coefficient, applying length penalties to reward model scores, and discarding overly long rollouts. Despite these efforts, RL-tuned models still tend to produce longer responses than SFT models. Recent work suggests that such verbosity does not always reflect deeper reasoning, but can also arise as an RL optimization artifact when models frequently produce incorrect answers, especially when training is dominated by unsolvable problems [42].

To address this, Kimi Team et al. [83] proposed several approaches to transform lengthy and unnecessary reasoning processes into concise and accurate ones. They investigated strategies like model merging, shortest rejection sampling, and DPO optimization for shorter responses. They also introduced the **long2short RL**, which uses a normalized reward model across multiple responses, significantly reducing output length while maintaining reasoning quality. The following section explores the long2short RL approach designed to improve the efficiency and effectiveness of RL-tuned LLMs.

Quality-Length Reward Disentangle. One approach to mitigate length bias is to develop more sophisticated reward models that can better distinguish between response quality and length [20, 134]. These methods employ a multi-head reward model on shared feature representations: one head is explicitly supervised to predict sequence length, while the other is optimized for quality. This architectural separation forces the model to isolate length information, ensuring the quality reward signal remains independent of the output length. Li et al. [90] propose the DRPO, decoupling the length-based optimization objectives for correct and incorrect rollouts, thereby preventing negative samples from interfering with the learning of positive ones. Investigating the correlation between accuracy and CoT length, Liang et al. [99] proposes an adaptive strategy that aligns reasoning depth with task complexity, encouraging concise responses for simple queries while reserving longer reasoning chains for difficult problems, thereby optimizing the efficiency-accuracy trade-off.

Length Penalty or Normalization. Meng et al. [121] proposed a simple length normalization way for DPO [146] to eliminate the influence of length, which has been proven quite effective in alleviating the length bias [24]. Yeo et al. [216] utilizes a cosine reward to incentivize different length scaling behaviors to regularize the model behavior, eliminating length bias. Chen et al. [24] observed that LRMs often provide multiple correct answers within a single solution. Leveraging this pattern, they split these solutions into multiple shorter ones and constructed preference pairs, with the first short and correct answer as the preferred solution. They optimized the LLMs using DPO and SimPO, finding SimPO [121] to be more effective, reducing response lengths by 30% to 40%. Beyond uniform length penalties, recent work proposes difficulty-aware penalties that encourage conciseness on easy questions while preserving sufficient reasoning depth on harder ones [105].

Procedure Reward. Rule-based reward [33], i.e., accuracy-based reward, has been proved to be quite effective, and less likely to suffer from reward hacking [160]. However, a recent study [144] highlights a limitation of accuracy-based rewards in shaping LLM reasoning: they fail to ensure efficient utilization of scaled computation during inference. To address this, the authors propose the Meta Reinforcement Finetuning (MRT) by using *cumulative regret* as a reinforcement learning objective, encouraging LLMs to make incremental progress with each token, to provide clear procedure instructions. This approach was shown to enable stable performance improvements as computational resources scale.

4.2.2 *Adaptive Budget-aware Tuning.*

Budget-aware tuning aims to align LRMs with explicit token budgets (e.g., reasoning trace length), enabling controllable trade-offs between accuracy and inference cost. Compared with purely test-time budget prediction and constrained generation (§ 5.1), this line of work primarily improves *intrinsic* budget following via post-training.

Budget Specification. A common starting point is to expose models to explicit budget signals at inference time, such as formulating the procedure as an instruction-following problem, by specifying a target length in the prompt [221]. More recent work conditions the model on more structured budget interfaces, e.g., inserting control tokens that indicate remaining budget during generation [190], or providing discrete user-controllable reasoning modes associated with different budget tiers [233].

Budget-aligned Post-training. To instill a more intrinsic awareness of resource limits, recent research has moved toward post-training models for budget compliance. While early efforts focused on supervised instruction-following with length constraints [221], subsequent works integrated RL to balance the trade-off between reasoning depth and compute budgets [4]. To address the challenge of capability collapse under tight budgets, Lyu et al. [114] proposed a hierarchical reward structure that preserves reasoning quality across different expenditure levels. Furthermore, researchers have introduced budget-aware architectural cues, such as remaining-budget tokens [190] or probability steering via Gamma distributions [92], to achieve finer-grained control. Above all, these advancements demonstrate that budget awareness can be effectively encoded into the model's parameters, transforming LRMs into resource-conscious ones that optimize the utility-to-cost ratio at the foundational level.

Adaptive Allocation and Gating. Since not all instances require the same reasoning depth, a key direction is to make budget usage adaptive. Yan et al. [204] trains models to allocate a reasoning quota conditioned on perceived difficulty, reducing unnecessary computation on easy queries while retaining depth for hard ones. Relatedly, hybrid-reasoning models learn to decide when to "think" and when to answer directly [74]. These approaches also help mitigate the token elasticity phenomenon, where overly strict constraints can trigger counterproductive behaviors and higher token costs [60], by favoring flexible, instance-dependent budget allocation over rigid length targets.

4.2.3 CoT Compression.

It is evident that longer CoT suffers from token redundancy and inference latency, which is not desired in practice. Thus, several studies try to identify important tokens and eliminate unnecessary tokens or reasoning steps, therefore enhancing the inference economy. These methods can be broadly categorized into two categories: 1) *explicit compression* that directly enforces the model to generate more concise reasoning by fine-tuning on carefully curated datasets or providing specific demonstrations.; and 2) *implicit compression* that maps multiple reasoning tokens or steps into a continuous space to achieve a more compact representation.

Explicit Compression. The first step of compression is to identify key tokens [86] or step in the reasoning processing, while most methods rely on perplexity as a primary metric, using it as a proxy to determine the importance of individual tokens or steps. Some of the methods refine the supervised fine-tuning dataset, replacing lengthy reasoning with identified key reasoning steps as target outputs [195]. Another approach uses key reasoning steps as demonstrations for in-context learning, guiding LLMs to generate only essential steps during inference while minimizing unimportant tokens [32]. A key limitation of these methods is the potential loss of coherence and a comprehensive understanding of the full reasoning process. While the full CoT can be reconstructed using additional LLMs, the recovered version may not fully align with the original, leading to inconsistencies or missing nuances and additional inference costs.

Implicit Compression. Besides explicit compression, another line of work argues that the explicit language space (i.e., tokens) may not always be optimal for reasoning, as most word tokens are primarily for textual coherence and not essential for reasoning [35, 61]. The earlier work focuses on the *implicit* chain-of-thought approach, which compiles explicit CoT reasoning of the teacher model into a student model that directly produces the final answer to develop more efficient

reasoning [34, 35]. Another representative work is Coconut (Chain of Continuous Thought) [61], which feed hidden state instead of specific token back to the LLM as the subsequent input embedding, allowing the model to encode multiple alternative next reasoning steps in the continuous space and thus reducing the token cost, followed by Cheng and Durme [27]. More recently, CoLaR introduces a latent head to dynamically compress reasoning chains in latent space and exposes a controllable compression factor at inference time [167]. Recent advances in memory-efficient architectures, such as Anchor-based LLMs (AnLLMs) [131], further highlight the inefficiency of token-based representations by demonstrating that compressing sequence information into anchor tokens can significantly reduce memory usage while maintaining accuracy, followed by Zhang et al. [228]. To reduce the need to fully finetune the models, recent studies use an additional projection module to inject the compressed continuous states of the source model into the target model [202].

4.3 Architecture

One approach to enhancing efficiency is reducing active model parameters, requiring adjustments in model or system architecture. Two key strategies are integrating System-1 and System-2 thinking and adaptively utilizing model depth. System-1 and System-2 cooperation enables dynamic selection between fast, intuitive reasoning and slower, deliberate processing, optimizing efficiency. Meanwhile, adaptive parameter activation optimizes model depth and resource allocation during inference, balancing performance and computational cost.

4.3.1 System-1 and System-2 Cooperation. Drawing inspiration from human cognition, several studies propose different methods to dynamically switch from slow, conscious, deliberate reasoning (System 2) to fast, automatic, intuitive thinking (System 1), therefore achieving a better balance between fast and accurate thought. Specifically, there are different paths towards this goal.

Single-Model Routing. It is straightforward to empower one model with abilities to switch between fast (system-1) and slow (system-2) inference methods according to difficulty signals, thereby optimizing both efficiency and effectiveness. On the one hand, the choice can be left to the user to determine the extent of cognitive effort exerted by the model. For instance, OpenAI has equipped the O1 [126] with three distinct thinking modes: low, middle, and high. The high thinking mode delivers accurate yet lengthy responses tailored for extremely challenging problems, while the low mode provides swift responses that are satisfactory. However, this direction has not been sufficiently explored thus far. How to train models to possess diverse modes of thinking and how to evaluate the capability gap between different modes [14] of thinking in models remain underexplored areas.

On the other hand, the model can be designed to autonomously determine the appropriate level of cognitive effort based on the complexity of the task at hand [38, 125, 130, 151, 179]. This autonomous selection process can be facilitated by incorporating a mechanism that assesses the difficulty of the input and dynamically adjusts the inference method accordingly. Ding et al. [38] leverages a binary router to assign queries to smaller or larger based on the difficulty and expected quality. Pan et al. [130] use consistency verification and reasoning steps as complexity signals to determine which questions require slow, deliberate thinking and which do not. Similarly, Saha et al. [151] curates a dataset that trains models to decompose problems into sub-goals of varying difficulty levels and adaptively apply different strategies to solve them.

Multi-Model Collaboration. One common approach to accelerating inference is speculative decoding [82, 87, 194], which follows a draft-then-verify paradigm. This method first generates multiple token candidates efficiently and then verifies them in parallel. Xia et al. [194] formally analyze this method through a formal study and extensive discussion

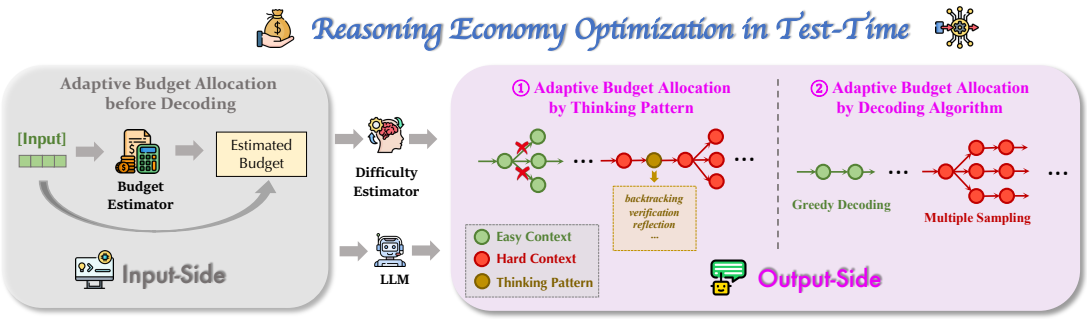


Fig. 6. Test-time Methods Optimization for Reasoning Economy. The methods are divided into adding an optimal computation constraint on the input-side, and selecting the best-performing decoding algorithm and controlling computation usage during decoding on the output-side.

of both drafting and verification phases, while Kim et al. [82] proposes the BiLD framework, where a smaller model generates text at low cost, and a larger model refines errors in parallel. Further research explores improved drafting and verification strategies using specific reward model [12, 100], with a broader overview in Xia et al. [196]. In addition, some methods take an agentic approach, dynamically determining when to engage larger models for refinement at different granularity [18, 103, 234]. Specifically, SwiftSage [103] invokes system 2 thinking for subgoals during inference, while CITER [234] focuses on token-level refinement when needed.

Knowledge Distillation. Another approach to achieving a balance between slow and fast reasoning is knowledge distillation, where knowledge from a larger, more complex model (System 2) is transferred to a smaller, more efficient model (System 1). The distilled smaller model will be capable of advanced reasoning abilities while costing less in computations. In detail, Yu et al. [219] leverages several self-supervised methods to distill higher-quality outputs from System 2 techniques back into LLM generations without intermediate reasoning token sequences. More recently, it has been proven that direct distillation from DeepSeek-R1 outperforms applying RL on the base model such as Qwen2.5-32B, setting a new record on the reasoning benchmarks [33]. There are also emerging studies exploring distillation from Transformer to lower computation complexity architectures [129], such as Mamba, achieving much faster inference for large batches and long sequences with the Mamba-based model.

4.3.2 *Adaptive Activated Parameters.*

Performance improvements of deep learning models largely benefit from advancements in deep network design [63]. Present LLMs also pay great attention to deep architecture designs. For models with comparable scales of parameters, increasing model depths often yields more significant benefits than broadening widths [25, 43], especially in mathematical reasoning tasks [214]. However, deepening layers will result in larger model sizes, which means that relatively small models with few layers are hard to resolve complex reasoning tasks. Additionally, increasing model layers leads to a linear increase in inference computation costs. From the perspective of model depth, we can enhance the inference capabilities of relatively small-scale LLMs by recurrently utilizing intermediate layers or accelerate the inference of deep LLMs by skipping some intermediate layers.

Recurrent Layers. Recurrent mechanisms enable LLMs to perform arbitrarily many computations before emitting a token, which is a simple solution for test-time compute scaling for relatively small LLMs with few layers [52]. Recent work has described depth-recurrent, looped Transformers and studied their potential benefits with careful theoretical and

small-scale analysis [50, 53, 208]. Specifically, Fan et al. [41] established the superior length generalization capabilities of looped Transformers. Yu et al. [220] proposes a framework enhancing CoT reasoning by combining looped and auto-regressive Transformers. Recurrently utilizing layers in Transformer-based LLMs enhances reasoning ability on small-scale LLMs.

Dynamic Depth. Additionally, inspired by efficient reasoning using model pruning and sparse models, recent works validate that not all layers of LLMs are necessary during inference [40]. In this way, two methods involving dynamic depths of LLMs are included: Early Exit [153, 171] and Skip Layer [80, 119, 211]. Since simpler tasks require fewer layers, while more complex tasks require more layers of inference, achieving efficient LLM inference then becomes when to stop the inference process adaptively based on the input instance [40].

5 Optimization for Reasoning Economy *part-2: Test-time Methods*

Test-time methods can enhance the performance of LRMs without the need for tuning. However, as previously noted (§ 3.2), there is significant potential for optimizing test-time usage to achieve greater reasoning efficiency. In this section, we survey existing methods aiming for reasoning economy by optimizing test-time methods to more economically allocate inference computation as shown in Figure 6.

5.1 Input-side Optimization

In this section, we introduce those works that attempt to optimize the usage of LLMs by controlling model behaviors through explicit constraints in the input.

5.1.1 Adaptive Budget Allocation before Decoding. The main idea of Adaptive Budget Allocation before Decoding is to decide a computation (thinking-token) budget for each query before generation and then steer the model to respect the budget during generation, enabling instance-level efficiency without changing the decoding algorithm [60, 177].

Budget Prediction. The basic idea of predicting the budget is to consider the difficulty of the problems to the reasoning LLM. According to previous works [16, 108, 203], the confidence of the model to solve the question can be estimated and predicted. As for the computation prediction methods for each question, Han et al. [60] collected a batch of questions with their optimal budgets and trained a regression model to predict the computation needed for novel prompts. Meanwhile, Wang et al. [177] considered the difficulty of the questions and allocated budgets similar to those of questions with comparable difficulty. Beyond external estimators, SelfBudgeter trains the model to explicitly self-estimate a suitable budget before producing the solution, and then optimizes for both budget adherence and efficiency via RL [97]. AdaCtrl exposes difficulty-aware budgeting through explicit tags, enabling both adaptive allocation and user override of the intended reasoning depth [69].

Budget Constrained Generation. Given a target budget, a straightforward approach is to specify length limits directly in the prompt [123, 221], which LLMs can follow to some extent [221]. L1 further improves controllability by using RL to optimize accuracy while adhering to user-specified length constraints [5]. Instead of hard constraints, Budget Guidance proposes a soft, token-level budget-conditioned generation method that guides generation with a lightweight predictor of remaining thinking length, improving accuracy under tight budgets without fine-tuning the base model [92].

5.2 Output-side Optimization

In this section, we introduce the work of optimizing the usage of LLMs at the output end of the model. This includes Constrained Decoding, Adaptive Algorithm Selection, and Adaptive Computation Allocation during Decoding.

5.2.1 Adaptive Algorithm Selection. The exploration of adaptive choice of test-time algorithms is quite limited, especially in efficient thinking. Here, we present several works that adaptively adjust the parameters of test-time algorithms, which could be potentially used to achieve algorithm determination.

Snell et al. [161] demonstrates the potential of adaptive method selection, showing that optimal settings can achieve up to 4x greater efficiency compared to the PRM best-of-N approach. This underscores the benefits of dynamically tailoring test-time strategies to specific tasks. Dhuliawala et al. [36] introduces an Adaptive Decoding Layer combined with Latent Preference Optimization (LPO), enabling fine-grained temperature prediction at the token or sample level for more precise generation control. Chen et al. [23] propose using high initial temperatures to enhance response diversity, which expands the search space and increases the likelihood of discovering correct answers. Wang et al. [178] further proposes an RL-based plugin that adaptively selects a factorized configuration (e.g., temperature, reasoning format, and reasoning steps) for each query, enabling more flexible test-time reasoning. These approaches collectively emphasize the importance of adaptive mechanisms in optimizing the choice of test-time algorithms, paving the way for more efficient and effective reasoning in LLMs.

5.2.2 Adaptive Budget Allocation During Decoding.

Early Stopping. The first line of adaptive budget allocation is early stopping, which aims to terminate generation once additional reasoning is unlikely to improve the final answer [3, 65, 96, 117, 172]. For example, early stopping can be driven by a model's self-evaluation signal during iterative self-refinement [117]. In parallel sampling settings such as self-consistency, a fixed sample size can be wasteful; thus, several methods stop sampling once consistency reaches a threshold [3] or progressively narrow the sampling window and terminate when answers converge within a window [96]. The same convergence principle has also been applied to single-trajectory CoT generation by monitoring provisional answers and exiting once they stabilize [109, 118].

Rather than using a uniform stopping rule, recent work emphasizes instance-dependent stopping criteria that account for both path quality and convergence [65, 172]. One representative approach combines confidence-based selection of preferred reasoning paths with consistency judgment to decide when to stop [65]. Beyond these heuristics, certainty-guided methods periodically probe the model during reasoning and stop once a target confidence threshold is reached [124], while internal-confidence based early-exit methods terminate at reasoning transition points when confidence becomes sufficiently high [206]. More principled signals can be derived from reasoning dynamics, e.g., detecting instance-specific "reasoning completion points" by monitoring termination-token rank dynamics [189]. Finally, REFRAIN formulates when-to-stop as an adaptive control problem, combining a redundancy discriminator with an online controller to adjust stopping thresholds under varying difficulty [166].

Search with Pruning. The second line is pruning while searching [142, 165, 185, 243], no matter the parallel or sequential methods. The goal is to prune low-quality search branches early while retaining high-quality ones, thereby saving computational resources. For sequential test-time methods, RMs are often used to guide the search procedure [175, 198, 217] by pruning low-quality branches and rolling out high-quality ones. Guided search could eliminate unnecessary computation usage while improving the accuracy of LRMs.

As for parallel test-time methods, Wang et al. [185] utilizes the self-evaluation abilities of LLMs to prune the low-quality samples during best-of-N sampling. To expand the search space of best-of-N, Sun et al. [165] set a very large sample window at the beginning of the sampling, large enough to consume all of the GPU memory, then gradually pruned those low-quality and completed the saved samples. Zhu et al. [243] select the top high-quality steps and use these steps to guide and constrain the LLMs to complete these prefixes.

Constrained Decoding. Identifying suboptimal thinking patterns is often challenging, which makes mitigating them particularly difficult. Hence, previous works have to utilize human analysis to diagnose the deceptive behaviors of LLMs [78, 182, 200] and more importantly, to utilize human observation to inspire specific solutions for them [176].

For example, for fake self-refine of LLMs, Huang et al. [66] and Gou et al. [54] suggest using external feedback to assist the self-refine procedure of LLMs. Other works utilize the human-observed behaviors to design force decoding paradigm, aiming to solve the fake thinking and overly cautious behaviors of LRMs, e.g., enforcing LLMs to adhere to the previous thought [182] or performing self-refine at proper time [122]. Wang et al. [182] found that the proposed Thought Switching Penalty approach can encourage LLMs to perform deeper reasoning and reduce unreasonable thought-switching.

6 Discussion

Efficient Multi-modal Reasoning. Multi-modal large language models (MLLMs) have demonstrated promising capabilities in various multi-modal reasoning tasks [181, 230], including but not limited to mathematical reasoning [111, 174, 241], visual question answering (VQA) [55, 72], and multi-modal dialogue systems [71]. Recent advances, exemplified by o1-like and R1-like technologies, have further catalyzed progress in this field [44, 70, 120, 135, 212, 227]. Current approaches on efficient multi-modal reasoning primarily focus on improvements within MLLMs themselves [77], mainly including: i) model architecture optimization (e.g., lightweight vision encoders [17], vision token compression [59], vision-language projector [91], small language models [29, 244]), and efficient structures [102, 104, 140]; ii) efficient vision techniques adoption (e.g. Pruning [218], knowledge distillation [170], and ViT quantization [95]), among other methods.

However, the evaluation and targeted optimization of efficiency in multi-modal (long-) reasoning remain relatively preliminary. Jiang et al. [73] proposes two metrics for measuring the efficiency of multi-model CoT: relevance rate, which calculates the proportion of content contributing to answering, and reflection quality, which evaluates whether reflection steps can effectively correct previous mistakes or present new insights. Xiang et al. [197] proposes a self-structured COT paradigm that decomposes reasoning into minimal semantic atomic steps to achieve efficient multi-modal reasoning by generating dynamic structures and lengths based on the problem types. Moreover, most of the aforementioned efficient reasoning methods for LLMs can also be effectively applied to MLLMs.

Efficient Agentic Reasoning. The advent of LRMs has also brought substantial performance improvements to AI agents. Using RL training similar to o1, the OpenAI's Deep Research [127] utilizes advanced reasoning capabilities to synthesize large amounts of online information, attaining notable performance on the challenging Humanity's Last Exam [136]. Recent research has increasingly focused on leveraging long reasoning capabilities in agent systems, where cutting-edge implementations integrate these with retrieval, tool augmentation, domain knowledge incorporation, and other auxiliary components to push performance boundaries [7, 49, 75].

To analyze the limited effectiveness of LRMs in interactive environments, Cuadron et al. [31] presents the first comprehensive empirical study in agentic tasks, analyzes the overthinking in LRMs, including its manifestations, quantification, and impact on different models, and also proposes potential mitigation approaches accordingly. Zhou

et al. [240] proposes the LaRMA framework to explore the necessity of reasoning capabilities in LRMs for agent scenarios, analyzes the performance differences between LRMs and LLMs across various tasks, and provides insights into optimizing agent performance through hybrid LLM-LRM configurations. Some studies also investigate efficient agent training methods [26] and efficient interaction schemes for multi-agent systems [186, 226, 239].

Evaluation Metrics and Benchmarks. With the proliferation of long reasoning and the increasing severity of the over-reasoning problem, researchers have recently begun developing specialized benchmarks and metrics to quantitatively measure reasoning efficiency. To stress-test overthinking and calibration, DNA Bench shows that current LRMs can generate up to 70× more unnecessary tokens and fail on tasks that non-reasoning models handle efficiently [62], while ThoughtTerminator introduces DUMB500 to evaluate calibration on extremely easy problems and proposes a training-free decoding method to improve it [138]. Beyond benchmarks, specialized metrics have also been proposed: Chen et al. [24] introduces outcome-based and process-based efficiency metrics and finds that LRMs often overthink on simple problems, with later reasoning contributing little to accuracy and diversity. Complementary to over-reasoning, underthinking has been quantified via an underthinking metric, revealing frequent premature thought switching in o1-like models that leads to insufficient reasoning depth [182]. Finally, Anderson et al. [8] builds a dataset from NPR Sunday Puzzle Challenges to evaluate reasoning with general (non-PhD-level) knowledge, which also surfaces overthinking-like behaviors.

Explainability of LRMs. The study of the explainability of black-box LLMs has always been a topic of interest [48, 51]. Particularly, LRMs have explored on their own through RL and have demonstrated reasoning abilities on par with those of human PhD students [33, 83, 126, 145]. However, the mechanisms underlying their achievement of such performance remain enigmatic. Current research on LRMs often focuses on their behavior analysis [31, 192, 216], such as the observation of overly cautious or fake thinking behaviors, and then retraces the post-training algorithms or test-time methods.

Nevertheless, it is essential to focus on how these models work on the inside, probing the internal mechanisms of LRMs [10, 11]. Moreover, more user-friendly and powerful toolkits [10, 149] are also critical for research purposes and large-scale human analysis. This would help us understand the thought patterns of LRMs, identify their faults, and provide directions for further improvement.

7 Conclusion

In this survey, we systematically examined the challenges and solutions involved in achieving an inference economy for large reasoning models, emphasizing the urgent need for efficient reasoning mechanisms that balance computational cost with performance. To the best of our knowledge, this is the first comprehensive review that analyzes the underlying causes, observed phenomena, key challenges, and emerging solutions for enabling efficient reasoning in LLMs, offering a structured roadmap and actionable strategies for practical deployment. By anchoring progress in the principles of inference economy, this survey serves not only as a synthesis of current knowledge but also as a call to action for further research in this path, highlighting the importance of developing more sustainable and scalable models that reason not only effectively but also efficiently.

References

[1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).

[2] Aradhye Agarwal, Ayan Sengupta, and Tanmoy Chakraborty. 2025. The Art of Scaling Test-Time Compute for Large Language Models. arXiv:2512.02008 [cs.CL] https://arxiv.org/abs/2512.02008

[3] Pranjal Aggarwal, Aman Madaan, Yiming Yang, and Mausam. 2023. Let's Sample Step by Step: Adaptive-Consistency for Efficient Reasoning and Coding with LLMs. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 12375–12396. doi:10.18653/v1/2023.emnlp-main.761

[4] Pranjal Aggarwal and Sean Welleck. 2025. L1: Controlling How Long A Reasoning Model Thinks With Reinforcement Learning. doi:10.48550/arXiv.2503.04697 arXiv:2503.04697 [cs].

[5] Pranjal Aggarwal and Sean Welleck. 2025. L1: Controlling How Long A Reasoning Model Thinks With Reinforcement Learning. In *Second Conference on Language Modeling*. https://openreview.net/forum?id=4jdIxXBNve

[6] AIME. 2025. Art of Problem Solving. https://artofproblemsolving.com/wiki/index.php/AIME_Problems_and_Solutions

[7] Salaheddin Alzubi, Creston Brooks, Purva Chiniya, Edoardo Contente, Chiara von Gerlach, Lucas Irwin, Yihan Jiang, Arda Kaz, Windsor Nguyen, Sewoong Oh, et al. 2025. Open Deep Search: Democratizing Search with Open-source Reasoning Agents. *arXiv preprint arXiv:2503.20201* (2025).

[8] Carolyn Jane Anderson, Joydeep Biswas, Aleksander Boruch-Gruszecki, Federico Cassano, Molly Q Feldman, Arjun Guha, Francesca Lucchetti, and Zixuan Wu. 2025. PhD Knowledge Not Required: A Reasoning Challenge for Large Language Models. arXiv:2502.01584 [cs.AI] https://arxiv.org/abs/2502.01584

[9] Zachary Ankner, Mansheej Paul, Brandon Cui, Jonathan D. Chang, and Prithviraj Ammanabrolu. 2024. Critique-out-Loud Reward Models. arXiv:2408.11791 [cs.LG] https://arxiv.org/abs/2408.11791

[10] Anthropic. 2025. Circuit Tracing: Revealing Computational Graphs in Language Models. https://transformer-circuits.pub/2025/attribution-graphs/methods.html

[11] Anthropic. 2025. On the Biology of a Large Language Model. https://transformer-circuits.pub/2025/attribution-graphs/biology.html

[12] Gregor Bachmann, Sotiris Anagnostidis, Albert Pumarola, Markos Georgopoulos, Artsiom Sanakoyeu, Yuming Du, Edgar Schönfeld, Ali Thabet, and Jonas Kohler. 2025. Judge Decoding: Faster Speculative Sampling Requires Going Beyond Model Alignment. *arXiv preprint arXiv:2501.19309* (2025).

[13] Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion, Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Tristan Hume, Scott Johnston, Shauna Kravec, Liane Lovitt, Neel Nanda, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, Ben Mann, and Jared Kaplan. 2022. Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback. arXiv:2204.05862 [cs.CL] https://arxiv.org/abs/2204.05862

[14] Marthe Ballon, Andres Algaba, and Vincent Ginis. 2025. The Relationship Between Reasoning and Performance in Large Language Models – o3 (mini) Thinks Harder, Not Longer. doi:10.48550/arXiv.2502.15631 arXiv:2502.15631 [cs].

[15] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. 2024. Large Language Monkeys: Scaling Inference Compute with Repeated Sampling. arXiv:2407.21787 [cs.LG] https://arxiv.org/abs/2407.21787

[16] Chao Chen, Kai Liu, Ze Chen, Yi Gu, Yue Wu, Mingyuan Tao, Zhihang Fu, and Jieping Ye. 2024. INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection. doi:10.48550/arXiv.2402.03744 arXiv:2402.03744.

[17] Jieneng Chen, Qihang Yu, Xiaohui Shen, Alan Yuille, and Liang-Chieh Chen. 2024. Vitamin: Designing scalable vision models in the vision-language era. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 12954–12966.

[18] Justin Chih-Yao Chen, Archiki Prasad, Swarnadeep Saha, Elias Stengel-Eskin, and Mohit Bansal. 2024. MAgICoRe: Multi-Agent, Iterative, Coarse-to-Fine Refinement for Reasoning. arXiv:2409.12147 [cs.CL] https://arxiv.org/abs/2409.12147

[19] Lingjiao Chen, Jared Quincy Davis, Boris Hanin, Peter Bailis, Ion Stoica, Matei Zaharia, and James Zou. 2024. Are More LLM Calls All You Need? Towards Scaling Laws of Compound Inference Systems. doi:10.48550/arXiv.2403.02419 arXiv:2403.02419.

[20] Lichang Chen, Chen Zhu, Jiuhai Chen, Davit Soselia, Tianyi Zhou, Tom Goldstein, Heng Huang, Mohammad Shoeybi, and Bryan Catanzaro. 2024. ODIN: Disentangled Reward Mitigates Hacking in RLHF. In *Forty-first International Conference on Machine Learning*. https://openreview.net/forum?id=zcIV8OQFVF

[21] Qiguang Chen, Libo Qin, Jinhao Liu, Dengyun Peng, Jiannan Guan, Peng Wang, Mengkang Hu, Yuhang Zhou, Te Gao, and Wanxiang Che. 2025. Towards Reasoning Era: A Survey of Long Chain-of-Thought for Reasoning Large Language Models. arXiv:2503.09567 [cs.AI] https://arxiv.org/abs/2503.09567

[22] Qiguang Chen, Libo Qin, Jiaqi Wang, Jingxuan Zhou, and Wanxiang Che. 2024. Unlocking the Capabilities of Thought: A Reasoning Boundary Framework to Quantify and Optimize Chain-of-Thought. https://openreview.net/forum?id=pC44UMwy2v

[23] Weizhe Chen, Zhicheng Zhang, Guanlin Liu, Renjie Zheng, Wenlei Shi, Chen Dun, Zheng Wu, Xing Jin, and Lin Yan. 2024. Flaming-hot Initiation with Regular Execution Sampling for Large Language Models. doi:10.48550/arXiv.2410.21236 arXiv:2410.21236.

[24] Xingyu Chen, Jiahao Xu, Tian Liang, Zhiwei He, Jianhui Pang, Dian Yu, Linfeng Song, Qiuzhi Liu, Mengfei Zhou, Zhuosheng Zhang, Rui Wang, Zhaopeng Tu, Haitao Mi, and Dong Yu. 2025. Do NOT Think That Much for 2+3=? On the Overthinking of o1-Like LLMs. arXiv:2412.21187 [cs.CL] https://arxiv.org/abs/2412.21187

[25] Xingwu Chen and Difan Zou. 2024. What can transformer learn with varying depth? case studies on sequence learning tasks. In *Proceedings of the 41st International Conference on Machine Learning* (Vienna, Austria) *(ICML'24)*. JMLR.org, Article 314, 30 pages.

[26] Zhixun Chen, Ming Li, Yuxuan Huang, Yali Du, Meng Fang, and Tianyi Zhou. 2025. ATLaS: Agent Tuning via Learning Critical Steps. *arXiv preprint arXiv:2503.02197* (2025).

[27] Jeffrey Cheng and Benjamin Van Durme. 2024. Compressed Chain of Thought: Efficient Reasoning Through Dense Representations. arXiv:2412.13171 [cs.CL] https://arxiv.org/abs/2412.13171

[28] Tianzhe Chu, Yuexiang Zhai, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V. Le, Sergey Levine, and Yi Ma. 2025. SFT Memorizes, RL Generalizes: A Comparative Study of Foundation Model Post-training. doi:10.48550/arXiv.2501.17161 arXiv:2501.17161 [cs].

[29] Xiangxiang Chu, Limeng Qiao, Xinyang Lin, Shuang Xu, Yang Yang, Yiming Hu, Fei Wei, Xinyu Zhang, Bo Zhang, Xiaolin Wei, et al. 2023. Mobilevlm: A fast, strong and open vision language assistant for mobile devices. *arXiv preprint arXiv:2312.16886* (2023).

[30] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training Verifiers to Solve Math Word Problems. arXiv:2110.14168 [cs.LG] https://arxiv.org/abs/2110.14168

[31] Alejandro Cuadron, Dacheng Li, Wenjie Ma, Xingyao Wang, Yichuan Wang, Siyuan Zhuang, Shu Liu, Luis Gaspar Schroeder, Tian Xia, Huanzhi Mao, Nicholas Thumiger, Aditya Desai, Ion Stoica, Ana Klimovic, Graham Neubig, and Joseph E. Gonzalez. 2025. The Danger of Overthinking: Examining the Reasoning-Action Dilemma in Agentic Tasks. doi:10.48550/arXiv.2502.08235 arXiv:2502.08235 [cs].

[32] Yingqian Cui, Pengfei He, Jingying Zeng, Hui Liu, Xianfeng Tang, Zhenwei Dai, Yan Han, Chen Luo, Jing Huang, Zhen Li, Suhang Wang, Yue Xing, Jiliang Tang, and Qi He. 2025. Stepwise Perplexity-Guided Refinement for Efficient Chain-of-Thought Reasoning in Large Language Models. arXiv:2502.13260 [cs.CL] https://arxiv.org/abs/2502.13260

[33] DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, and Xiao et al. Bi. 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. doi:10.48550/arXiv.2501.12948 arXiv:2501.12948 [cs].

[34] Yuntian Deng, Yejin Choi, and Stuart Shieber. 2024. From Explicit CoT to Implicit CoT: Learning to Internalize CoT Step by Step. arXiv:2405.14838 [cs.CL] https://arxiv.org/abs/2405.14838

[35] Yuntian Deng, Kiran Prasad, Roland Fernandez, Paul Smolensky, Vishrav Chaudhary, and Stuart Shieber. 2023. Implicit Chain of Thought Reasoning via Knowledge Distillation. arXiv:2311.01460 [cs.CL] https://arxiv.org/abs/2311.01460

[36] Shehzaad Dhuliawala, Ilia Kulikov, Ping Yu, Asli Celikyilmaz, Jason Weston, Sainbayar Sukhbaatar, and Jack Lanchantin. 2024. Adaptive Decoding via Latent Preference Optimization. http://arxiv.org/abs/2411.09661 arXiv:2411.09661 version: 1.

[37] Raul Cavalcante Dinardi, Bruno Yamamoto, Anna Helena Reali Costa, and Artur Jordao. 2025. The Virtues of Brevity: Avoid Overthinking in Parallel Test-Time Reasoning. In *NeurIPS 2025 Workshop on Efficient Reasoning*. https://openreview.net/forum?id=giq139C56f

[38] Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Ruhle, Laks V. S. Lakshmanan, and Ahmed Hassan Awadallah. 2024. Hybrid LLM: Cost-Efficient and Quality-Aware Query Routing. doi:10.48550/arXiv.2404.14618

[39] Alexander R Fabbri, Wojciech Kryściński, Bryan McCann, Caiming Xiong, Richard Socher, and Dragomir Radev. 2021. Summeval: Re-evaluating summarization evaluation. *Transactions of the Association for Computational Linguistics* 9 (2021), 391–409.

[40] Siqi Fan, Xin Jiang, Xiang Li, Xuying Meng, Peng Han, Shuo Shang, Aixin Sun, Yequan Wang, and Zhongyuan Wang. 2024. Not All Layers of LLMs Are Necessary During Inference. arXiv:2403.02181 [cs.CL] https://arxiv.org/abs/2403.02181

[41] Ying Fan, Yilun Du, Kannan Ramchandran, and Kangwook Lee. 2025. Looped Transformers for Length Generalization. In *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=2edigk8yoU

[42] Mehdi Fatemi, Banafsheh Rafiee, Mingjie Tang, and Kartik Talamadupula. 2025. Concise Reasoning via Reinforcement Learning. arXiv:2504.05185 [cs.CL] https://arxiv.org/abs/2504.05185

[43] Guhao Feng, Bohang Zhang, Yuntian Gu, Haotian Ye, Di He, and Liwei Wang. 2023. Towards Revealing the Mystery behind Chain of Thought: A Theoretical Perspective. In *Thirty-seventh Conference on Neural Information Processing Systems*. https://openreview.net/forum?id=qHrADgAdYu

[44] Kaituo Feng, Kaixiong Gong, Bohao Li, Zonghao Guo, Yibing Wang, Tianshuo Peng, Benyou Wang, and Xiangyu Yue. 2025. Video-R1: Reinforcing Video Reasoning in MLLMs. *arXiv preprint arXiv:2503.21776* (2025).

[45] Sicheng Feng, Gongfan Fang, Xinyin Ma, and Xinchao Wang. 2025. Efficient Reasoning Models: A Survey. arXiv:2504.10903 [cs.CL] https://arxiv.org/abs/2504.10903

[46] Yunlong Feng, Yang Xu, Libo Qin, Yasheng Wang, and Wanxiang Che. 2024. Improving Language Model Reasoning with Self-motivated Learning. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, Nicoletta Calzolari, Min-Yen Kan, Veronique Hoste, Alessandro Lenci, Sakriani Sakti, and Nianwen Xue (Eds.). ELRA and ICCL, Torino, Italia, 8840–8852. https://aclanthology.org/2024.lrec-main.774/

[47] Leo Gao, John Schulman, and Jacob Hilton. 2023. Scaling laws for reward model overoptimization. In *Proceedings of the 40th International Conference on Machine Learning* (Honolulu, Hawaii, USA) *(ICML'23)*. JMLR.org, Article 437, 32 pages.

[48] Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. 2024. Scaling and evaluating sparse autoencoders. doi:10.48550/arXiv.2406.04093 arXiv:2406.04093 [cs].

[49] Shanghua Gao, Richard Zhu, Zhenglun Kong, Ayush Noori, Xiaorui Su, Curtis Ginder, Theodoros Tsiligkaridis, and Marinka Zitnik. 2025. TxAgent: An AI Agent for Therapeutic Reasoning Across a Universe of Tools. *arXiv preprint arXiv:2503.10970* (2025).

[50] Khashayar Gatmiry, Nikunj Saunshi, Sashank J. Reddi, Stefanie Jegelka, and Sanjiv Kumar. 2024. Can Looped Transformers Learn to Implement Multi-step Gradient Descent for In-context Learning? arXiv:2410.08292 [cs.LG] https://arxiv.org/abs/2410.08292

[51] Xuyang Ge, Fukang Zhu, Wentao Shu, Junxuan Wang, Zhengfu He, and Xipeng Qiu. 2024. Automatically Identifying Local and Global Circuits with Linear Computation Graphs. doi:10.48550/arXiv.2405.13868 arXiv:2405.13868 [cs].

[52] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. 2025. Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach. arXiv:2502.05171 [cs.LG] https://arxiv.org/abs/2502.05171

[53] Angeliki Giannou, Shashank Rajput, Jy-Yong Sohn, Kangwook Lee, Jason D. Lee, and Dimitris Papailiopoulos. 2023. Looped Transformers as Programmable Computers. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 202)*, Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (Eds.). PMLR, 11398–11442. https://proceedings.mlr.press/v202/giannou23a.html

[54] Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. [n. d.]. CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing. ([n. d.]).

[55] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2017. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 6904–6913.

[56] Ryan Greenblatt, Carson Denison, Benjamin Wright, Fabien Roger, Monte MacDiarmid, Sam Marks, Johannes Treutlein, Tim Belonax, Jack Chen, David Duvenaud, Akbir Khan, Julian Michael, Sören Mindermann, Ethan Perez, Linda Petrini, Jonathan Uesato, Jared Kaplan, Buck Shlegeris, Samuel R. Bowman, and Evan Hubinger. 2024. Alignment faking in large language models. arXiv:2412.14093 [cs.AI] https://arxiv.org/abs/2412.14093

[57] Caglar Gulcehre, Tom Le Paine, Srivatsan Srinivasan, Ksenia Konyushkova, Lotte Weerts, Abhishek Sharma, Aditya Siddhant, Alex Ahern, Miaosen Wang, Chenjie Gu, et al. 2023. Reinforced self-training (rest) for language modeling. *arXiv preprint arXiv:2308.08998* (2023).

[58] Jiani Guo, Zuchao Li, Jie Wu, Qianren Wang, Yun Li, Lefei Zhang, Hai Zhao, and Yujiu Yang. 2025. ToM: Leveraging Tree-oriented MapReduce for Long-Context Reasoning in Large Language Models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (Eds.). Association for Computational Linguistics, Suzhou, China, 17793–17812. doi:10.18653/v1/2025.emnlp-main.899

[59] Zonghao Guo, Ruyi Xu, Yuan Yao, Junbo Cui, Zanlin Ni, Chunjiang Ge, Tat-Seng Chua, Zhiyuan Liu, and Gao Huang. 2024. Llava-uhd: an lmm perceiving any aspect ratio and high-resolution images. In *European Conference on Computer Vision*. Springer, 390–406.

[60] Tingxu Han, Zhenting Wang, Chunrong Fang, Shiyu Zhao, Shiqing Ma, and Zhenyu Chen. 2025. Token-Budget-Aware LLM Reasoning. doi:10.48550/arXiv.2412.18547 arXiv:2412.18547 [cs].

[61] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. 2024. Training Large Language Models to Reason in a Continuous Latent Space. arXiv:2412.06769 [cs.CL] https://arxiv.org/abs/2412.06769

[62] Masoud Hashemi, Oluwanifemi Bamgbose, Sathwik Tejaswi Madhusudhan, Jishnu Sethumadhavan Nair, Aman Tiwari, and Vikas Yadav. 2025. DNA Bench: When Silence is Smarter – Benchmarking Over-Reasoning in Reasoning LLMs. doi:10.48550/arXiv.2503.15793 arXiv:2503.15793 [cs].

[63] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

[64] Andreas Hochlehnert, Hardik Bhatnagar, Vishaal Udandarao, Samuel Albanie, Ameya Prabhu, and Matthias Bethge. 2025. A Sober Look at Progress in Language Model Reasoning: Pitfalls and Paths to Reproducibility. arXiv:2504.07086 [cs.LG] https://arxiv.org/abs/2504.07086

[65] Chengsong Huang, Langlin Huang, Jixuan Leng, Jiacheng Liu, and Jiaxin Huang. 2025. Efficient Test-Time Scaling via Self-Calibration. doi:10.48550/arXiv.2503.00031 arXiv:2503.00031 [cs].

[66] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. 2023. Large Language Models Cannot Self-Correct Reasoning Yet. https://openreview.net/forum?id=IkmD3fKBPQ

[67] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. 2024. Large Language Models Cannot Self-Correct Reasoning Yet. In *The Twelfth International Conference on Learning Representations*. https://openreview.net/forum?id=IkmD3fKBPQ

[68] Jiaxin Huang, Shixiang Gu, Le Hou, Yuexin Wu, Xuezhi Wang, Hongkun Yu, and Jiawei Han. 2023. Large Language Models Can Self-Improve. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 1051–1068. doi:10.18653/v1/2023.emnlp-main.67

[69] Shijue Huang, Hongru Wang, Wanjun Zhong, Zhaochen Su, Jiazhan Feng, Bowen Cao, and Yi R. Fung. 2025. AdaCtrl: Towards Adaptive and Controllable Reasoning via Difficulty-Aware Budgeting. arXiv:2505.18822 [cs.AI] https://arxiv.org/abs/2505.18822

[70] Wenxuan Huang, Bohan Jia, Zijie Zhai, Shaosheng Cao, Zheyu Ye, Fei Zhao, Yao Hu, and Shaohui Lin. 2025. Vision-r1: Incentivizing reasoning capability in multimodal large language models. *arXiv preprint arXiv:2503.06749* (2025).

[71] Yupan Huang, Zaiqiao Meng, Fangyu Liu, Yixuan Su, Nigel Collier, and Yutong Lu. 2023. Sparkles: Unlocking chats across multiple images for multimodal instruction-following models. *arXiv preprint arXiv:2308.16463* (2023).

[72] Drew A Hudson and Christopher D Manning. 2019. Gqa: A new dataset for real-world visual reasoning and compositional question answering. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 6700–6709.

[73] Dongzhi Jiang, Renrui Zhang, Ziyu Guo, Yanwei Li, Yu Qi, Xinyan Chen, Liuhui Wang, Jianhan Jin, Claire Guo, Shen Yan, et al. 2025. MME-CoT: Benchmarking Chain-of-Thought in Large Multimodal Models for Reasoning Quality, Robustness, and Efficiency. *arXiv preprint arXiv:2502.09621* (2025).

[74] Lingjie Jiang, Xun Wu, Shaohan Huang, Qingxiu Dong, Zewen Chi, Li Dong, Xingxing Zhang, Tengchao Lv, Lei Cui, and Furu Wei. 2025. Think Only When You Need with Large Hybrid-Reasoning Models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.

https://openreview.net/forum?id=fDjDVE4qdj

[75] Bowen Jin, Hansi Zeng, Zhenrui Yue, Dong Wang, Hamed Zamani, and Jiawei Han. 2025. Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning. *arXiv preprint arXiv:2503.09516* (2025).

[76] Mingyu Jin, Qinkai Yu, Dong Shu, Haiyan Zhao, Wenyue Hua, Yanda Meng, Yongfeng Zhang, and Mengnan Du. 2024. The Impact of Reasoning Step Length on Large Language Models. In *Findings of the Association for Computational Linguistics: ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 1830–1842. doi:10.18653/v1/2024.findings-acl.108

[77] Yizhang Jin, Jian Li, Yexin Liu, Tianjun Gu, Kai Wu, Zhengkai Jiang, Muyang He, Bo Zhao, Xin Tan, Zhenye Gan, et al. 2024. Efficient multimodal large language models: A survey. *arXiv preprint arXiv:2405.10739* (2024).

[78] Ryo Kamoi, Yusen Zhang, Nan Zhang, Jiawei Han, and Rui Zhang. 2024. When Can LLMs Actually Correct Their Own Mistakes? A Critical Survey of Self-Correction of LLMs. *Transactions of the Association for Computational Linguistics* 12 (2024), 1417–1440. doi:10.1162/tacl_a_00713

[79] Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T Joshi, Hanna Moazam, et al. 2023. Dspy: Compiling declarative language model calls into self-improving pipelines. *arXiv preprint arXiv:2310.03714* (2023).

[80] Bo-Kyeong Kim, Geonmin Kim, Tae-Ho Kim, Thibault Castells, Shinkook Choi, Junho Shin, and Hyoung-Kyu Song. 2024. Shortened LLaMA: Depth Pruning for Large Language Models with Comparison of Retraining Methods. arXiv:2402.02834 [cs.LG] https://arxiv.org/abs/2402.02834

[81] Jeonghye Kim, Sojeong Rhee, Minbeom Kim, Dohyung Kim, Sangmook Lee, Youngchul Sung, and Kyomin Jung. 2025. ReflAct: World-Grounded Decision Making in LLM Agents via Goal-State Reflection. arXiv:2505.15182 [cs.CL] https://arxiv.org/abs/2505.15182

[82] Sehoon Kim, Karttikeya Mangalam, Suhong Moon, Jitendra Malik, Michael W Mahoney, Amir Gholami, and Kurt Keutzer. 2023. Speculative decoding with big little decoder. *Advances in Neural Information Processing Systems* 36 (2023), 39236–39256.

[83] Angang Kimi Team, and Du, Bofei Gao, Bowei Xing, Changjiu Jiang, Cheng Chen, Cheng Li, Chenjun Xiao, Chenzhuang Du, Chonghua Liao, Chuning Tang, Congcong Wang, Dehao Zhang, Enming Yuan, Enzhe Lu, Fengxiang Tang, Flood Sung, Guangda Wei, Guokun Lai, Haiqing Guo, Han Zhu, Hao Ding, Hao Hu, Hao Yang, Hao Zhang, Haotian Yao, Haotian Zhao, Haoyu Lu, Haoze Li, Haozhen Yu, Hongcheng Gao, Huabin Zheng, Huan Yuan, Jia Chen, Jianhang Guo, Jianlin Su, Jianzhou Wang, Jie Zhao, Jin Zhang, Jingyuan Liu, Junjie Yan, Junyan Wu, Lidong Shi, Ling Ye, Longhui Yu, Mengnan Dong, Neo Zhang, Ningchen Ma, Qiwei Pan, Qucheng Gong, Shaowei Liu, Shengling Ma, Shupeng Wei, Sihan Cao, Siying Huang, Tao Jiang, Weihao Gao, Weimin Xiong, Weiran He, Weixiao Huang, Wenhao Wu, Wenyang He, Xianghui Wei, Xianqing Jia, Xingzhe Wu, Xinran Xu, Xinxing Zu, Xinyu Zhou, Xuehai Pan, Y. Charles, Yang Li, Yangyang Hu, Yangyang Liu, Yanru Chen, Yejie Wang, Yibo Liu, Yidao Qin, Yifeng Liu, Ying Yang, Yiping Bao, Yulun Du, Yuxin Wu, Yuzhi Wang, Zaida Zhou, Zhaoji Wang, Zhaowei Li, Zhen Zhu, Zheng Zhang, Zhexu Wang, Zhilin Yang, Zhiqi Huang, Zihao Huang, Ziyao Xu, and Zonghan Yang. 2025. Kimi k1.5: Scaling Reinforcement Learning with LLMs. doi:10.48550/arXiv.2501.12599 arXiv:2501.12599 [cs].

[84] Aviral Kumar, Vincent Zhuang, Rishabh Agarwal, Yi Su, John D. Co-Reyes, Avi Singh, Kate Baumli, Shariq Iqbal, Colton Bishop, Rebecca Roelofs, Lei M. Zhang, Kay McKinney, Disha Shrivastava, Cosmin Paduraru, George Tucker, Doina Precup, Feryal Behbahani, and Aleksandra Faust. 2024. Training Language Models to Self-Correct via Reinforcement Learning. http://arxiv.org/abs/2409.12917 arXiv:2409.12917.

[85] Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. 2019. Natural Questions: A Benchmark for Question Answering Research. *Transactions of the Association for Computational Linguistics* 7 (2019), 452–466. doi:10.1162/tacl_a_00276

[86] Ayeong Lee, Ethan Che, and Tianyi Peng. 2025. How Well do LLMs Compress Their Own Chain-of-Thought? A Token Complexity Approach. doi:10.48550/arXiv.2503.01141 arXiv:2503.01141 [cs].

[87] Yaniv Leviathan, Matan Kalman, and Yossi Matias. 2023. Fast inference from transformers via speculative decoding. In *International Conference on Machine Learning*. PMLR, 19274–19286.

[88] Mosh Levy, Alon Jacoby, and Yoav Goldberg. 2024. Same Task, More Tokens: the Impact of Input Length on the Reasoning Performance of Large Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 15339–15353. doi:10.18653/v1/2024.acl-long.818

[89] Mosh Levy, Alon Jacoby, and Yoav Goldberg. 2024. Same Task, More Tokens: the Impact of Input Length on the Reasoning Performance of Large Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 15339–15353. doi:10.18653/v1/2024.acl-long.818

[90] Gang Li, Yan Chen, Ming Lin, and Tianbao Yang. 2025. DRPO: Efficient Reasoning via Decoupled Reward Policy Optimization. arXiv:2510.04474 [cs.AI] https://arxiv.org/abs/2510.04474

[91] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*. PMLR, 19730–19742.

[92] Junyan Li, Wenshuo Zhao, Yang Zhang, and Chuang Gan. 2025. Steering LLM Thinking with Budget Guidance. arXiv:2506.13752 [cs.CL] https://arxiv.org/abs/2506.13752

[93] Siheng Li, Cheng Yang, Zesen Cheng, Lemao Liu, Mo Yu, Yujiu Yang, and Wai Lam. 2024. Large Language Models Can Self-Improve in Long-context Reasoning. *arXiv preprint arXiv:2411.08147* (2024).

[94] Xuefeng Li, Haoyang Zou, and Pengfei Liu. 2025. LIMR: Less is More for RL Scaling. arXiv:2502.11886 [cs.LG] https://arxiv.org/abs/2502.11886

[95] Yanjing Li, Sheng Xu, Baochang Zhang, Xianbin Cao, Peng Gao, and Guodong Guo. 2022. Q-vit: Accurate and fully quantized low-bit vision transformer. *Advances in neural information processing systems* 35 (2022), 34451–34463.

[96] Yiwei Li, Peiwen Yuan, Shaoxiong Feng, Boyuan Pan, Xinglin Wang, Bin Sun, Heda Wang, and Kan Li. 2024. Escape Sky-high Cost: Early-stopping Self-Consistency for Multi-step Reasoning. arXiv:2401.10480 [cs.CL] https://arxiv.org/abs/2401.10480

[97] Zheng Li, Qingxiu Dong, Jingyuan Ma, Di Zhang, Kai Jia, and Zhifang Sui. 2026. SelfBudgeter: Adaptive Token Allocation for Efficient LLM Reasoning. arXiv:2505.11274 [cs.AI] https://arxiv.org/abs/2505.11274

[98] Zhong-Zhi Li, Duzhen Zhang, Ming-Liang Zhang, Jiaxin Zhang, Zengyan Liu, Yuxuan Yao, Haotian Xu, Junhao Zheng, Pei-Jie Wang, Xiuyi Chen, Yingying Zhang, Fei Yin, Jiahua Dong, Zhijiang Guo, Le Song, and Cheng-Lin Liu. 2025. From System 1 to System 2: A Survey of Reasoning Large Language Models. doi:10.48550/arXiv.2502.17419 arXiv:2502.17419 [cs].

[99] Tian Liang, Wenxiang Jiao, Zhiwei He, Jiahao Xu, Haitao Mi, and Dong Yu. 2025. DeepCompress: A Dual Reward Strategy for Dynamically Exploring and Compressing Reasoning Chains. arXiv:2510.27419 [cs.AI] https://arxiv.org/abs/2510.27419

[100] Baohao Liao, Yuhui Xu, Hanze Dong, Junnan Li, Christof Monz, Silvio Savarese, Doyen Sahoo, and Caiming Xiong. 2025. Reward-Guided Speculative Decoding for Efficient LLM Reasoning. arXiv:2501.19324 [cs.CL] https://arxiv.org/abs/2501.19324

[101] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2023. Let's Verify Step by Step. http://arxiv.org/abs/2305.20050 arXiv:2305.20050.

[102] Bin Lin, Zhenyu Tang, Yang Ye, Jiaxi Cui, Bin Zhu, Peng Jin, Jinfa Huang, Junwu Zhang, Yatian Pang, Munan Ning, et al. 2024. Moe-llava: Mixture of experts for large vision-language models. *arXiv preprint arXiv:2401.15947* (2024).

[103] Bill Yuchen Lin, Yicheng Fu, Karina Yang, Faeze Brahman, Shiyu Huang, Chandra Bhagavatula, Prithviraj Ammanabrolu, Yejin Choi, and Xiang Ren. 2023. Swiftsage: A generative agent with fast and slow thinking for complex interactive tasks. *Advances in Neural Information Processing Systems* 36 (2023), 23813–23825.

[104] Zhihang Lin, Mingbao Lin, Luxi Lin, and Rongrong Ji. 2024. Boosting multimodal large language models with visual tokens withdrawal for rapid inference. *arXiv preprint arXiv:2405.05803* (2024).

[105] Zehui Ling, Deshu Chen, Hongwei Zhang, Yifeng Jiao, Xin Guo, and Yuan Cheng. 2025. Fast on the Easy, Deep on the Hard: Efficient Reasoning via Powered Length Penalty. arXiv:2506.10446 [cs.CL] https://arxiv.org/abs/2506.10446

[106] Jiacheng Liu, Andrew Cohen, Ramakanth Pasunuru, Yejin Choi, Hannaneh Hajishirzi, and Asli Celikyilmaz. 2024. Don't throw away your value model! Generating more preferable text with Value-Guided Monte-Carlo Tree Search decoding. arXiv:2309.15028 [cs.CL] https://arxiv.org/abs/2309.15028

[107] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics* 12 (2024), 157–173. doi:10.1162/tacl_a_00638 Place: Cambridge, MA Publisher: MIT Press.

[108] Shudong Liu, Zhaocong Li, Xuebo Liu, Runzhe Zhan, Derek F. Wong, Lidia S. Chao, and Min Zhang. 2024. Can LLMs Learn Uncertainty on Their Own? Expressing Uncertainty Effectively in A Self-Training Manner. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Miami, Florida, USA, 21635–21645. https://aclanthology.org/2024.emnlp-main.1205

[109] Xin Liu and Lu Wang. 2025. Answer Convergence as a Signal for Early Stopping in Reasoning. arXiv:2506.02536 [cs.CL] https://arxiv.org/abs/2506.02536

[110] Junru Lu, Jiazheng Li, Siyu An, Meng Zhao, Yulan He, Di Yin, and Xing Sun. 2024. Eliminating Biased Length Reliance of Direct Preference Optimization via Down-Sampled KL Divergence. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 1047–1067. doi:10.18653/v1/2024.emnlp-main.60

[111] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. 2024. MathVista: Evaluating Mathematical Reasoning of Foundation Models in Visual Contexts. In *International Conference on Learning Representations (ICLR)*.

[112] Liangchen Luo, Yinxiao Liu, Rosanne Liu, Samrat Phatale, Harsh Lara, Yunxuan Li, Lei Shu, Yun Zhu, Lei Meng, Jiao Sun, et al. 2024. Improve mathematical reasoning in language models by automated process supervision. *arXiv preprint arXiv:2406.06592* 2 (2024).

[113] Trung Quoc Luong, Xinbo Zhang, Zhanming Jie, Peng Sun, Xiaoran Jin, and Hang Li. 2024. ReFT: Reasoning with Reinforced Fine-Tuning. arXiv:2401.08967 [cs.CL] https://arxiv.org/abs/2401.08967

[114] Shangke Lyu, Linjuan Wu, Yuchen Yan, Xingyu Wu, Hao Li, Yongliang Shen, Peisheng Jiang, Weiming Lu, Jun Xiao, and Yueting Zhuang. 2025. Hierarchical Budget Policy Optimization for Adaptive Reasoning. arXiv:2507.15844 [cs.AI] https://arxiv.org/abs/2507.15844

[115] Xinyin Ma, Gongfan Fang, and Xinchao Wang. 2023. LLM-Pruner: On the Structural Pruning of Large Language Models. arXiv:2305.11627 [cs.CL] https://arxiv.org/abs/2305.11627

[116] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. SELF-REFINE: iterative refinement with self-feedback. In *Proceedings of the 37th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) *(NIPS '23)*. Curran Associates Inc., Red Hook, NY, USA, Article 2019, 61 pages.

[117] Rohin Manvi, Anikait Singh, and Stefano Ermon. 2024. Adaptive Inference-Time Compute: LLMs Can Predict if They Can Do Better, Even Mid-Generation. arXiv:2410.02725 [cs.CL] https://arxiv.org/abs/2410.02725

[118] Minjia Mao, Bowen Yin, Yu Zhu, and Xiao Fang. 2025. Early Stopping Chain-of-thoughts in Large Language Models. arXiv:2509.14004 [cs.CL] https://arxiv.org/abs/2509.14004

[119] Xin Men, Mingyu Xu, Qingyu Zhang, Bingning Wang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. 2024. ShortGPT: Layers in Large Language Models are More Redundant Than You Expect. arXiv:2403.03853 [cs.CL] https://arxiv.org/abs/2403.03853

[120] Fanqing Meng, Lingxiao Du, Zongkai Liu, Zhixiang Zhou, Quanfeng Lu, Daocheng Fu, Botian Shi, Wenhai Wang, Junjun He, Kaipeng Zhang, et al. 2025. MM-Eureka: Exploring Visual Aha Moment with Rule-based Large-scale Reinforcement Learning. *arXiv preprint arXiv:2503.07365* (2025).

[121] Yu Meng, Mengzhou Xia, and Danqi Chen. 2024. SimPO: Simple Preference Optimization with a Reference-Free Reward. doi:10.48550/arXiv.2405.14734 arXiv:2405.14734 [cs].

[122] Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori Hashimoto. 2025. s1: Simple test-time scaling. arXiv:2501.19393 [cs.CL] https://arxiv.org/abs/2501.19393

[123] Sania Nayab, Giulio Rossolini, Marco Simoni, Andrea Saracino, Giorgio Buttazzo, Nicolamaria Manes, and Fabrizio Giacomelli. 2025. Concise Thoughts: Impact of Output Length on LLM Reasoning and Cost. arXiv:2407.19825 [cs.CL] https://arxiv.org/abs/2407.19825

[124] João Paulo Nogueira, Wentao Sun, Alonso Silva, and Laith Zumot. 2025. Certainty-Guided Reasoning in Large Language Models: A Dynamic Thinking Budget Approach. arXiv:2509.07820 [cs.AI] https://arxiv.org/abs/2509.07820

[125] Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M Waleed Kadous, and Ion Stoica. 2025. RouteLLM: Learning to Route LLMs with Preference Data. arXiv:2406.18665 [cs.LG] https://arxiv.org/abs/2406.18665

[126] OpenAI. 2024. OpenAI o1. https://openai.com/o1

[127] OpenAI. 2025. Deep Research. https://openai.com/index/introducing-deep-research

[128] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Proceedings of the 36th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) *(NIPS '22)*. Curran Associates Inc., Red Hook, NY, USA, Article 2011, 15 pages.

[129] Daniele Paliotta, Junxiong Wang, Matteo Pagliardini, Kevin Y Li, Aviv Bick, J Zico Kolter, Albert Gu, François Fleuret, and Tri Dao. 2025. Thinking Slow, Fast: Scaling Inference Compute with Distilled Reasoners. *arXiv preprint arXiv:2502.20339* (2025).

[130] Jiabao Pan, Yan Zhang, Chen Zhang, Zuozhu Liu, Hongwei Wang, and Haizhou Li. 2024. DynaThink: Fast or Slow? A Dynamic Decision-Making Framework for Large Language Models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 14686–14695. doi:10.18653/v1/2024.emnlp-main.814

[131] Jianhui Pang, Fanghua Ye, Derek Wong, Xin He, Wanshun Chen, and Longyue Wang. 2024. Anchor-based Large Language Models. In *Findings of the Association for Computational Linguistics: ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 4958–4976. doi:10.18653/v1/2024.findings-acl.295

[132] Jianhui Pang, Fanghua Ye, Derek Fai Wong, Dian Yu, Shuming Shi, Zhaopeng Tu, and Longyue Wang. 2025. Salute the classic: Revisiting challenges of machine translation in the age of large language models. *Transactions of the Association for Computational Linguistics* 13 (2025), 73–95.

[133] Shubham Parashar, Blake Olson, Sambhav Khurana, Eric Li, Hongyi Ling, James Caverlee, and Shuiwang Ji. 2025. Inference-Time Computations for LLM Reasoning and Planning: A Benchmark and Insights. arXiv:2502.12521 [cs.AI] https://arxiv.org/abs/2502.12521

[134] Ryan Park, Rafael Rafailov, Stefano Ermon, and Chelsea Finn. 2024. Disentangling Length from Quality in Direct Preference Optimization. In *Findings of the Association for Computational Linguistics: ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 4998–5017. doi:10.18653/v1/2024.findings-acl.297

[135] Yingzhe Peng, Gongrui Zhang, Miaosen Zhang, Zhiyuan You, Jie Liu, Qipeng Zhu, Kai Yang, Xingzhong Xu, Xin Geng, and Xu Yang. 2025. LMM-R1: Empowering 3B LMMs with Strong Reasoning Abilities Through Two-Stage Rule-Based RL. *arXiv preprint arXiv:2503.07536* (2025).

[136] Long Phan, Alice Gatti, Ziwen Han, Nathaniel Li, Josephina Hu, Hugh Zhang, Chen Bo Calvin Zhang, Mohamed Shaaban, John Ling, Sean Shi, et al. 2025. Humanity's Last Exam. *arXiv preprint arXiv:2501.14249* (2025).

[137] Xiao Pu, Michael Saxon, Wenyue Hua, and William Yang Wang. 2025. THOUGHTTERMINATOR: Benchmarking, Calibrating, and Mitigating Overthinking in Reasoning Models. arXiv:2504.13367 [cs.CL] https://arxiv.org/abs/2504.13367

[138] Xiao Pu, Michael Saxon, Wenyue Hua, and William Yang Wang. 2025. ThoughtTerminator: Benchmarking, Calibrating, and Mitigating Overthinking in Reasoning Models. In *Second Conference on Language Modeling*. https://openreview.net/forum?id=oHR862dpMC

[139] Jianing Qi, Xi Ye, Hao Tang, Zhigang Zhu, and Eunsol Choi. 2025. Learning to Reason Across Parallel Samples for LLM Reasoning. In *NeurIPS 2025 Workshop on Efficient Reasoning*. https://openreview.net/forum?id=lVjZPH9Zyq

[140] Yanyuan Qiao, Zheng Yu, Longteng Guo, Sihan Chen, Zijia Zhao, Mingzhen Sun, Qi Wu, and Jing Liu. 2024. Vl-mamba: Exploring state space models for multimodal learning. *arXiv preprint arXiv:2403.13600* (2024).

[141] Yiwei Qin, Xuefeng Li, Haoyang Zou, Yixiu Liu, Shijie Xia, Zhen Huang, Yixin Ye, Weizhe Yuan, Hector Liu, Yuanzhi Li, and Pengfei Liu. 2024. O1 Replication Journey: A Strategic Progress Report – Part 1. doi:10.48550/arXiv.2410.18982 arXiv:2410.18982 [cs].

[142] Jiahao Qiu, Yifu Lu, Yifan Zeng, Jiacheng Guo, Jiayi Geng, Huazheng Wang, Kaixuan Huang, Yue Wu, and Mengdi Wang. 2024. Treebon: Enhancing inference-time alignment with speculative tree-search and best-of-n sampling. *arXiv preprint arXiv:2410.16033* (2024).

[143] Xiaoye Qu, Yafu Li, Zhao-Chen Su, Weigao Sun, Jianhao Yan, Dongrui Liu, Ganqu Cui, Daizong Liu, Shuxian Liang, Junxian He, Peng Li, Wei Wei, Jing Shao, Chaochao Lu, Yue Zhang, Xian-Sheng Hua, Bowen Zhou, and Yu Cheng. 2025. A Survey of Efficient Reasoning for Large Reasoning Models: Language, Multimodality, and Beyond. arXiv:2503.21614 [cs.CL] https://arxiv.org/abs/2503.21614

[144] Yuxiao Qu, Matthew Y. R. Yang, Amrith Setlur, Lewis Tunstall, Edward Emanuel Beeching, Ruslan Salakhutdinov, and Aviral Kumar. 2025. Optimizing Test-Time Compute via Meta Reinforcement Fine-Tuning. http://arxiv.org/abs/2503.07572 arXiv:2503.07572.

[145] QwQ. 2025. QwQ-32B: Embracing the Power of Reinforcement Learning | Qwen. https://qwenlm.github.io/blog/qwq-32b/

[146] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. In *Advances in Neural Information Processing Systems*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (Eds.), Vol. 36. Curran Associates, Inc., 53728–53741. https://proceedings.neurips.cc/paper_files/paper/2023/file/a85b405ed65c6477a4fe8302b5e06ce7-Paper-Conference.pdf

[147] Hossein A. Rahmani, Satyapriya Krishna, Xi Wang, Mohammadmehdi Naghiaei, and Emine Yilmaz. 2025. Self-Correcting Large Language Models: Generation vs. Multiple Choice. arXiv:2511.09381 [cs.CL] https://arxiv.org/abs/2511.09381

[148] Shyam Sundhar Ramesh, Yifan Hu, Iason Chaimalas, Viraj Mehta, Pier Giuseppe Sessa, Haitham Bou Ammar, and Ilija Bogunovic. 2024. Group Robust Preference Optimization in Reward-free RLHF. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*. https://openreview.net/forum?id=PRAsjrmXXK

[149] Anton Razzhigaev, Matvey Mikhalchuk, Temurbek Rahmatullaev, Elizaveta Goncharova, Polina Druzhinina, Ivan Oseledets, and Andrey Kuznetsov. 2025. LLM-Microscope: Uncovering the Hidden Role of Punctuation in Context Memory of Transformers. doi:10.48550/arXiv.2502.15007 arXiv:2502.15007 [cs].

[150] Zhiwen Ruan, Yixia Li, Yefeng Liu, Yun Chen, Weihua Luo, Peng Li, Yang Liu, and Guanhua Chen. 2025. G2: Guided Generation for Enhanced Output Diversity in LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (Eds.). Association for Computational Linguistics, Suzhou, China, 14116–14134. doi:10.18653/v1/2025.emnlp-main.713

[151] Swarnadeep Saha, Archiki Prasad, Justin Chih-Yao Chen, Peter Hase, Elias Stengel-Eskin, and Mohit Bansal. 2024. System-1. x: Learning to balance fast and slow planning with language models. *arXiv preprint arXiv:2407.14414* (2024).

[152] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal Policy Optimization Algorithms. arXiv:1707.06347 [cs.LG]

[153] Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Tran, Yi Tay, and Donald Metzler. 2022. Confident Adaptive Language Modeling. In *Advances in Neural Information Processing Systems*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh (Eds.), Vol. 35. Curran Associates, Inc., 17456–17472. https://proceedings.neurips.cc/paper_files/paper/2022/file/6fac9e316a4ae75ea244ddcef1982c71-Paper-Conference.pdf

[154] Amrith Setlur, Nived Rajaraman, Sergey Levine, and Aviral Kumar. 2025. Scaling Test-Time Compute Without Verification or RL is Suboptimal. doi:10.48550/arXiv.2502.12118 arXiv:2502.12118 [cs].

[155] Xu Shen, Song Wang, Zhen Tan, Laura Yao, Xinyu Zhao, Kaidi Xu, Xin Wang, and Tianlong Chen. 2025. FaithCoT-Bench: Benchmarking Instance-Level Faithfulness of Chain-of-Thought Reasoning. arXiv:2510.04040 [cs.AI] https://arxiv.org/abs/2510.04040

[156] Haizhou Shi, Ye Liu, Bo Pang, Zeyu Leo Liu, Hao Wang, Silvio Savarese, Caiming Xiong, Yingbo Zhou, and Semih Yavuz. 2025. SSR: Socratic Self-Refine for Large Language Model Reasoning. arXiv:2511.10621 [cs.CL] https://arxiv.org/abs/2511.10621

[157] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik R Narasimhan, and Shunyu Yao. 2023. Reflexion: language agents with verbal reinforcement learning. In *Thirty-seventh Conference on Neural Information Processing Systems*. https://openreview.net/forum?id=vAElhFcKW6

[158] Prasann Singhal, Tanya Goyal, Jiacheng Xu, and Greg Durrett. 2024. A Long Way to Go: Investigating Length Correlations in RLHF. In *First Conference on Language Modeling*. https://openreview.net/forum?id=G8LaO1P0xv

[159] Nishad Singhi, Hritik Bansal, Arian Hosseini, Aditya Grover, Kai-Wei Chang, Marcus Rohrbach, and Anna Rohrbach. 2025. When To Solve, When To Verify: Compute-Optimal Problem Solving and Generative Verification for LLM Reasoning. arXiv:2504.01005 [cs.CL] https://arxiv.org/abs/2504.01005

[160] Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. 2022. Defining and Characterizing Reward Hacking. arXiv:2209.13085 [cs.LG] https://arxiv.org/abs/2209.13085

[161] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. 2024. Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters. doi:10.48550/arXiv.2408.03314 arXiv:2408.03314 [cs].

[162] Nisan Stiennon, Long Ouyang, Jeff Wu, Daniel M. Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul Christiano. 2020. Learning to summarize from human feedback. In *Proceedings of the 34th International Conference on Neural Information Processing Systems (NIPS '20)*. Curran Associates Inc., Red Hook, NY, USA, 3008–3021.

[163] Vighnesh Subramaniam, Yilun Du, Joshua B Tenenbaum, Antonio Torralba, Shuang Li, and Igor Mordatch. 2025. Multiagent finetuning: Self improvement with diverse reasoning chains. *arXiv preprint arXiv:2501.05707* (2025).

[164] Yang Sui, Yu-Neng Chuang, Guanchu Wang, Jiamu Zhang, Tianyi Zhang, Jiayi Yuan, Hongyi Liu, Andrew Wen, Shaochen Zhong, Na Zou, Hanjie Chen, and Xia Hu. 2025. Stop Overthinking: A Survey on Efficient Reasoning for Large Language Models. arXiv:2503.16419 [cs.CL] https://arxiv.org/abs/2503.16419

[165] Hanshi Sun, Momin Haider, Ruiqi Zhang, Huitao Yang, Jiahao Qiu, Ming Yin, Mengdi Wang, Peter Bartlett, and Andrea Zanette. 2024. Fast Best-of-N Decoding via Speculative Rejection. doi:10.48550/arXiv.2410.20290 arXiv:2410.20290 [cs].

[166] Renliang Sun, Wei Cheng, Dawei Li, Haifeng Chen, and Wei Wang. 2025. Stop When Enough: Adaptive Early-Stopping for Chain-of-Thought Reasoning. arXiv:2510.10103 [cs.CL] https://arxiv.org/abs/2510.10103

[167] Wenhui Tan, Jiaze Li, Jianzhong Ju, Zhenbo Luo, Jian Luan, and Ruihua Song. 2025. Think Silently, Think Fast: Dynamic Latent Compression of LLM Reasoning Chains. arXiv:2505.16552 [cs.CL] https://arxiv.org/abs/2505.16552

[168] Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. 2023. Stanford Alpaca: An Instruction-following LLaMA model. https://github.com/tatsu-lab/stanford_alpaca.

[169] Guiyao Tie, Zenghui Yuan, Zeli Zhao, Chaoran Hu, Tianhe Gu, Ruihang Zhang, Sizhe Zhang, Junran Wu, Xiaoyue Tu, Ming Jin, Qingsong Wen, Lixing Chen, Pan Zhou, and Lichao Sun. 2025. Can LLMs Correct Themselves? A Benchmark of Self-Correction in LLMs. arXiv:2510.16062 [cs.CL] https://arxiv.org/abs/2510.16062

[170] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. 2021. Training data-efficient image transformers & distillation through attention. In *International conference on machine learning*. PMLR, 10347–10357.

[171] Neeraj Varshney, Agneet Chatterjee, Mihir Parmar, and Chitta Baral. 2024. Investigating Acceleration of LLaMA Inference by Enabling Intermediate Layer Decoding via Instruction Tuning with 'LITE'. In *Findings of the Association for Computational Linguistics: NAACL 2024*, Kevin Duh, Helena Gomez, and Steven Bethard (Eds.). Association for Computational Linguistics, Mexico City, Mexico, 3656–3677. doi:10.18653/v1/2024.findings-naacl.232

[172] Guangya Wan, Yuqi Wu, Jie Chen, and Sheng Li. 2025. Reasoning Aware Self-Consistency: Leveraging Reasoning Paths for Efficient LLM Sampling. doi:10.48550/arXiv.2408.17017 arXiv:2408.17017 [cs].

[173] Hongru Wang, Deng Cai, Wanjun Zhong, Shijue Huang, Jeff Z Pan, Zeming Liu, and Kam-Fai Wong. 2025. Self-Reasoning Language Models: Unfold Hidden Reasoning Chains with Few Reasoning Catalyst. In *Workshop on Reasoning and Planning for Large Language Models*.

[174] Ke Wang, Junting Pan, Weikang Shi, Zimu Lu, Houxing Ren, Aojun Zhou, Mingjie Zhan, and Hongsheng Li. 2024. Measuring multimodal mathematical reasoning with math-vision dataset. *Advances in Neural Information Processing Systems* 37 (2024), 95095–95169.

[175] Peiyi Wang, Lei Li, Zhihong Shao, R. X. Xu, Damai Dai, Yifei Li, Deli Chen, Y. Wu, and Zhifang Sui. 2024. Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations. doi:10.48550/arXiv.2312.08935 arXiv:2312.08935.

[176] Rui Wang, Hongru Wang, Fei Mi, Boyang Xue, Yi Chen, Kam-Fai Wong, and Ruifeng Xu. 2024. Enhancing Large Language Models Against Inductive Instructions with Dual-critique Prompting. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Kevin Duh, Helena Gomez, and Steven Bethard (Eds.). Association for Computational Linguistics, Mexico City, Mexico, 5345–5363. doi:10.18653/v1/2024.naacl-long.299

[177] Xinglin Wang, Shaoxiong Feng, Yiwei Li, Peiwen Yuan, Yueqi Zhang, Chuyi Tan, Boyuan Pan, Yao Hu, and Kan Li. 2025. Make Every Penny Count: Difficulty-Adaptive Self-Consistency for Cost-Efficient Reasoning. arXiv:2408.13457 [cs.CL] https://arxiv.org/abs/2408.13457

[178] Xiangqi Wang, Yue Huang, Yanbo Wang, Xiaonan Luo, Kehan Guo, Yujun Zhou, and Xiangliang Zhang. 2025. AdaReasoner: Adaptive Reasoning Enables More Flexible Thinking in Large Language Models. arXiv:2505.17312 [cs.AI] https://arxiv.org/abs/2505.17312

[179] Xinyuan Wang, Yanchi Liu, Wei Cheng, Xujiang Zhao, Zhengzhang Chen, Wenchao Yu, Yanjie Fu, and Haifeng Chen. 2025. MixLLM: Dynamic Routing in Mixed Large Language Models. arXiv:2502.18482 [cs.CL] https://arxiv.org/abs/2502.18482

[180] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. In *The Eleventh International Conference on Learning Representations*. https://openreview.net/forum?id=1PL1NIMMrw

[181] Yiqi Wang, Wentao Chen, Xiaotian Han, Xudong Lin, Haiteng Zhao, Yongfei Liu, Bohan Zhai, Jianbo Yuan, Quanzeng You, and Hongxia Yang. 2024. Exploring the reasoning abilities of multimodal large language models (mllms): A comprehensive survey on emerging trends in multimodal reasoning. *arXiv preprint arXiv:2401.06805* (2024).

[182] Yue Wang, Qiuzhi Liu, Jiahao Xu, Tian Liang, Xingyu Chen, Zhiwei He, Linfeng Song, Dian Yu, Juntao Li, Zhuosheng Zhang, Rui Wang, Zhaopeng Tu, Haitao Mi, and Dong Yu. 2025. Thoughts Are All Over the Place: On the Underthinking of o1-Like LLMs. doi:10.48550/arXiv.2501.18585 arXiv:2501.18585 [cs].

[183] Yixu Wang, Yan Teng, Kexin Huang, Chengqi Lyu, Songyang Zhang, Wenwei Zhang, Xingjun Ma, Yu-Gang Jiang, Yu Qiao, and Yingchun Wang. 2024. Fake Alignment: Are LLMs Really Aligned Well?. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Kevin Duh, Helena Gomez, and Steven Bethard (Eds.). Association for Computational Linguistics, Mexico City, Mexico, 4696–4712. doi:10.18653/v1/2024.naacl-long.263

[184] Yiping Wang, Qing Yang, Zhiyuan Zeng, Liliang Ren, Liyuan Liu, Baolin Peng, Hao Cheng, Xuehai He, Kuan Wang, Jianfeng Gao, Weizhu Chen, Shuohang Wang, Simon Shaolei Du, and Yelong Shen. 2025. Reinforcement Learning for Reasoning in Large Language Models with One Training Example. arXiv:2504.20571 [cs.LG] https://arxiv.org/abs/2504.20571

[185] Yiming Wang, Pei Zhang, Siyuan Huang, Baosong Yang, Zhuosheng Zhang, Fei Huang, and Rui Wang. 2025. Sampling-Efficient Test-Time Scaling: Self-Estimating the Best-of-N Sampling in Early Decoding. doi:10.48550/arXiv.2503.01422 arXiv:2503.01422 [cs].

[186] Zhexuan Wang, Yutong Wang, Xuebo Liu, Liang Ding, Miao Zhang, Jie Liu, and Min Zhang. 2025. AgentDropout: Dynamic Agent Elimination for Token-Efficient and High-Performance LLM-Based Multi-Agent Collaboration. *arXiv preprint arXiv:2503.18891* (2025).

[187] Jason Wei, Maarten Bosma, Vincent Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M Dai, and Quoc V Le. 2021. Finetuned Language Models are Zero-Shot Learners. In *International Conference on Learning Representations*.

[188] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Proceedings of the 36th International Conference on Neural Information Processing Systems (NIPS '22)*. Curran Associates Inc., Red Hook, NY, USA, 24824–24837.

[189] Zihao Wei, Liang Pang, Jiahao Liu, Wenjie Shi, Jingcheng Deng, Shicheng Xu, Zenghao Duan, Fei Sun, Huawei Shen, and Xueqi Cheng. 2026. The Evolution of Thought: Tracking LLM Overthinking via Reasoning Dynamics Analysis. arXiv:2508.17627 [cs.CL] https://arxiv.org/abs/2508.17627

[190] Hao Wen, Xinrui Wu, Yi Sun, Feifei Zhang, Liye Chen, Jie Wang, Yunxin Liu, Yunhao Liu, Ya-Qin Zhang, and Yuanchun Li. 2025. BudgetThinker: Empowering Budget-aware LLM Reasoning with Control Tokens. arXiv:2508.17196 [cs.LG] https://arxiv.org/abs/2508.17196

[191] Wanxing Wu, He Zhu, Yixia Li, Lei Yang, Jiehui Zhao, Hongru Wang, Jian Yang, Benyou Wang, Bingyi Jing, and Guanhua Chen. 2026. Towards Fair and Comprehensive Evaluation of Routers in Collaborative LLM Systems. arXiv:2602.11877 [cs.CL] https://arxiv.org/abs/2602.11877

[192] Yuyang Wu, Yifei Wang, Tianqi Du, Stefanie Jegelka, and Yisen Wang. 2025. When More is Less: Understanding Chain-of-Thought Length in LLMs. doi:10.48550/arXiv.2502.07266 arXiv:2502.07266 [cs].

[193] Zeqiu Wu, Yushi Hu, Weijia Shi, Nouha Dziri, Alane Suhr, Prithviraj Ammanabrolu, Noah A. Smith, Mari Ostendorf, and Hannaneh Hajishirzi. 2023. Fine-grained human feedback gives better rewards for language model training. In *Proceedings of the 37th International Conference on Neural Information Processing Systems (NIPS '23)*. Curran Associates Inc., Red Hook, NY, USA, 59008–59033.

[194] Heming Xia, Tao Ge, Peiyi Wang, Si-Qing Chen, Furu Wei, and Zhifang Sui. 2023. Speculative Decoding: Exploiting Speculative Execution for Accelerating Seq2seq Generation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 3909–3925. doi:10.18653/v1/2023.findings-emnlp.257

[195] Heming Xia, Yongqi Li, Chak Tou Leong, Wenjie Wang, and Wenjie Li. 2025. TokenSkip: Controllable Chain-of-Thought Compression in LLMs. arXiv:2502.12067 [cs.CL] https://arxiv.org/abs/2502.12067

[196] Heming Xia, Zhe Yang, Qingxiu Dong, Peiyi Wang, Yongqi Li, Tao Ge, Tianyu Liu, Wenjie Li, and Zhifang Sui. 2024. Unlocking Efficiency in Large Language Model Inference: A Comprehensive Survey of Speculative Decoding. In *Findings of the Association for Computational Linguistics: ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 7655–7671. doi:10.18653/v1/2024.findings-acl.456

[197] Kun Xiang, Zhili Liu, Zihao Jiang, Yunshuang Nie, Kaixin Cai, Yiyang Yin, Runhui Huang, Haoxiang Fan, Hanhui Li, Weiran Huang, et al. 2025. Can Atomic Step Decomposition Enhance the Self-structured Reasoning of Multimodal Large Models? *arXiv preprint arXiv:2503.06252* (2025).

[198] Yuxi Xie, Kenji Kawaguchi, Yiran Zhao, Xu Zhao, Min-Yen Kan, Junxian He, and Qizhe Xie. 2023. Self-Evaluation Guided Beam Search for Reasoning. arXiv:2305.00633 [cs.CL] https://arxiv.org/abs/2305.00633

[199] Huajian Xin, Daya Guo, Zhihong Shao, Zhizhou Ren, Qihao Zhu, Bo Liu, Chong Ruan, Wenda Li, and Xiaodan Liang. 2024. DeepSeek-Prover: Advancing Theorem Proving in LLMs through Large-Scale Synthetic Data. arXiv:2405.14333 [cs.AI] https://arxiv.org/abs/2405.14333

[200] Ruoxi Xu, Hongyu Lin, Xianpei Han, Jia Zheng, Weixiang Zhou, Le Sun, and Yingfei Sun. 2024. Large Language Models Often Say One Thing and Do Another. https://openreview.net/forum?id=RTHbao4Mib

[201] Xin Xu, Tianhao Chen, Fan Zhang, Wanlong Liu, Pengxiang Li, Ajay Kumar Jaiswal, Yuchen Yan, Jishan Hu, Yang Wang, Hao Chen, Shiwei Liu, Shizhe Diao, Can Yang, and Lu Yin. 2025. Double-Checker: Enhancing Reasoning of Slow-Thinking LLMs via Self-Critical Fine-Tuning. arXiv:2506.21285 [cs.CL] https://arxiv.org/abs/2506.21285

[202] Yige Xu, Xu Guo, Zhiwei Zeng, and Chunyan Miao. 2025. SoftCoT: Soft Chain-of-Thought for Efficient Reasoning with LLMs. arXiv:2502.12134 [cs.CL] https://arxiv.org/abs/2502.12134

[203] Boyang Xue, Fei Mi, Qi Zhu, Hongru Wang, Rui Wang, Sheng Wang, Erxin Yu, Xuming Hu, and Kam-Fai Wong. 2024. UAlign: Leveraging Uncertainty Estimations for Factuality Alignment on Large Language Models. doi:10.48550/arXiv.2412.11803 arXiv:2412.11803 [cs].

[204] Kaiwen Yan, Xuanqing Shi, Hongcheng Guo, Wenxuan Wang, Zhuosheng Zhang, and Chengwei Qin. 2025. DRQA: Dynamic Reasoning Quota Allocation for Controlling Overthinking in Reasoning Large Language Models. arXiv:2508.17803 [cs.CL] https://arxiv.org/abs/2508.17803

[205] An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, Keming Lu, Mingfeng Xue, Runji Lin, Tianyu Liu, Xingzhang Ren, and Zhenru Zhang. 2024. Qwen2.5-Math Technical Report: Toward Mathematical Expert Model via Self-Improvement. arXiv:2409.12122 [cs.CL] https://arxiv.org/abs/2409.12122

[206] Chenxu Yang, Qingyi Si, Yongjie Duan, Zheliang Zhu, Chenyu Zhu, Qiaowei Li, Minghui Chen, Zheng Lin, and Weiping Wang. 2025. Dynamic Early Exit in Reasoning Models. arXiv:2504.15895 [cs.CL] https://arxiv.org/abs/2504.15895

[207] Kevin Yang, Dan Klein, Asli Celikyilmaz, Nanyun Peng, and Yuandong Tian. 2024. RLCD: Reinforcement Learning from Contrastive Distillation for LM Alignment. In *The Twelfth International Conference on Learning Representations*. https://openreview.net/forum?id=v3XXtxWKi6

[208] Liu Yang, Kangwook Lee, Robert D Nowak, and Dimitris Papailiopoulos. 2024. Looped Transformers are Better at Learning Learning Algorithms. In *The Twelfth International Conference on Learning Representations*. https://openreview.net/forum?id=HHbRxoDTxE

[209] Ling Yang, Zhaochen Yu, Tianjun Zhang, Shiyi Cao, Minkai Xu, Wentao Zhang, Joseph E. Gonzalez, and Bin CUI. 2024. Buffer of Thoughts: Thought-Augmented Reasoning with Large Language Models. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*. https://openreview.net/forum?id=ANO1i9JPtb

[210] Wenkai Yang, Shuming Ma, Yankai Lin, and Furu Wei. 2025. Towards Thinking-Optimal Scaling of Test-Time Compute for LLM Reasoning. arXiv:2502.18080 [cs.CL] https://arxiv.org/abs/2502.18080

[211] Yifei Yang, Zouying Cao, and Hai Zhao. 2024. LaCo: Large Language Model Pruning via Layer Collapse. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 6401–6417. doi:10.18653/v1/2024.findings-emnlp.372

[212] Huanjin Yao, Jiaxing Huang, Wenhao Wu, Jingyi Zhang, Yibo Wang, Shunyu Liu, Yingjie Wang, Yuxin Song, Haocheng Feng, Li Shen, et al. 2024. Mulberry: Empowering mllm with o1-like reasoning and reflection via collective monte carlo tree search. *arXiv preprint arXiv:2412.18319* (2024).

[213] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of Thoughts: Deliberate Problem Solving with Large Language Models. doi:10.48550/arXiv.2305.10601 arXiv:2305.10601 [cs].

[214] Tian Ye, Zicheng Xu, Yuanzhi Li, and Zeyuan Allen-Zhu. 2025. Physics of Language Models: Part 2.1, Grade-School Math and the Hidden Reasoning Process. In *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=Tn5B6Udq3E

[215] Yixin Ye, Zhen Huang, Yang Xiao, Ethan Chern, Shijie Xia, and Pengfei Liu. 2025. LIMO: Less is More for Reasoning. arXiv:2502.03387 [cs.CL] https://arxiv.org/abs/2502.03387

[216] Edward Yeo, Yuxuan Tong, Morry Niu, Graham Neubig, and Xiang Yue. 2025. Demystifying Long Chain-of-Thought Reasoning in LLMs. doi:10.48550/arXiv.2502.03373 arXiv:2502.03373 [cs].

[217] Fei Yu, Anningzhe Gao, and Benyou Wang. 2024. OVM, Outcome-supervised Value Models for Planning in Mathematical Reasoning. In *Findings of the Association for Computational Linguistics: NAACL 2024*, Kevin Duh, Helena Gomez, and Steven Bethard (Eds.). Association for Computational Linguistics, Mexico City, Mexico, 858–875. doi:10.18653/v1/2024.findings-naacl.55

[218] Fang Yu, Kun Huang, Meng Wang, Yuan Cheng, Wei Chu, and Li Cui. 2022. Width & depth pruning for vision transformers. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 3143–3151.

[219] Ping Yu, Jing Xu, Jason Weston, and Ilia Kulikov. 2024. Distilling system 2 into system 1. *arXiv preprint arXiv:2407.06023* (2024).

[220] Qifan Yu, Zhenyu He, Sijie Li, Xun Zhou, Jun Zhang, Jingjing Xu, and Di He. 2025. Enhancing Auto-regressive Chain-of-Thought through Loop-Aligned Reasoning. arXiv:2502.08482 [cs.CL] https://arxiv.org/abs/2502.08482

[221] Weizhe Yuan, Ilia Kulikov, Ping Yu, Kyunghyun Cho, Sainbayar Sukhbaatar, Jason Weston, and Jing Xu. 2024. Following Length Constraints in Instructions. arXiv:2406.17744 [cs.CL] https://arxiv.org/abs/2406.17744

[222] Linan Yue, Yichao Du, Yizhi Wang, Weibo Gao, Fangzhou Yao, Li Wang, Ye Liu, Ziyu Xu, Qi Liu, Shimin Di, and Min-Ling Zhang. 2025. Don't Overthink It: A Survey of Efficient R1-style Large Reasoning Models. arXiv:2508.02120 [cs.AI] https://arxiv.org/abs/2508.02120

[223] Eric Zelikman, Georges Raif Harik, Yijia Shao, Varuna Jayasiri, Nick Haber, and Noah Goodman. 2024. Quiet-star: Language models can teach themselves to think before speaking. In *First Conference on Language Modeling*.

[224] Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah Goodman. 2022. Star: Bootstrapping reasoning with reasoning. *Advances in Neural Information Processing Systems* 35 (2022), 15476–15488.

[225] Yongcheng Zeng, Xinyu Cui, Xuanfa Jin, Qirui Mi, Guoqing Liu, Zexu Sun, Mengyue Yang, Dong Li, Weiyu Ma, Ning Yang, Jian Zhao, Jianye Hao, Haifeng Zhang, and Jun Wang. 2025. Evolving LLMs' Self-Refinement Capability via Synergistic Training-Inference Optimization. arXiv:2502.05605 [cs.CL] https://arxiv.org/abs/2502.05605

[226] Guibin Zhang, Yanwei Yue, Zhixun Li, Sukwon Yun, Guancheng Wan, Kun Wang, Dawei Cheng, Jeffrey Xu Yu, and Tianlong Chen. 2025. Cut the Crap: An Economical Communication Pipeline for LLM-based Multi-Agent Systems. In *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=LkzuPorQ5L

[227] Jingyi Zhang, Jiaxing Huang, Huanjin Yao, Shunyu Liu, Xikun Zhang, Shijian Lu, and Dacheng Tao. 2025. R1-VL: Learning to Reason with Multimodal Large Language Models via Step-wise Group Relative Policy Optimization. *arXiv preprint arXiv:2503.12937* (2025).

[228] Jintian Zhang, Yuqi Zhu, Mengshu Sun, Yujie Luo, Shuofei Qiao, Lun Du, Da Zheng, Huajun Chen, and Ningyu Zhang. 2025. LightThinker: Thinking Step-by-Step Compression. arXiv:2502.15589 [cs.CL] https://arxiv.org/abs/2502.15589

[229] Yanzhi Zhang, Yitong Duan, Zhaoxi Zhang, Jiyan He, and Shuxin Zheng. 2025. Population-Evolve: a Parallel Sampling and Evolutionary Method for LLM Math Reasoning. arXiv:2512.19081 [cs.AI] https://arxiv.org/abs/2512.19081

[230] Zhuosheng Zhang, Aston Zhang, Mu Li, George Karypis, Alex Smola, et al. 2024. Multimodal Chain-of-Thought Reasoning in Language Models. *Transactions on Machine Learning Research* (2024).

[231] Zhenru Zhang, Chujie Zheng, Yangzhen Wu, Beichen Zhang, Runji Lin, Bowen Yu, Dayiheng Liu, Jingren Zhou, and Junyang Lin. 2025. The lessons of developing process reward models in mathematical reasoning. *arXiv preprint arXiv:2501.07301* (2025).

[232] Zhenru Zhang, Chujie Zheng, Yangzhen Wu, Beichen Zhang, Runji Lin, Bowen Yu, Dayiheng Liu, Jingren Zhou, and Junyang Lin. 2025. The Lessons of Developing Process Reward Models in Mathematical Reasoning. arXiv:2501.07301 [cs.CL] https://arxiv.org/abs/2501.07301

[233] Kai Zhao, Yanjun Zhao, Jiaming Song, Shien He, Lusheng Zhang, Qiang Zhang, and Tianjiao Li. 2025. SABER: Switchable and Balanced Training for Efficient LLM Reasoning. arXiv:2508.10026 [cs.CL] https://arxiv.org/abs/2508.10026

[234] Wenhao Zheng, Yixiao Chen, Weitong Zhang, Souvik Kundu, Yun Li, Zhengzhong Liu, Eric P Xing, Hongyi Wang, and Huaxiu Yao. 2025. CITER: Collaborative Inference for Efficient Large Language Model Decoding with Token-Level Routing. *arXiv preprint arXiv:2502.01976* (2025).

[235] Wanjun Zhong, Ruixiang Cui, Yiduo Guo, Yaobo Liang, Shuai Lu, Yanlin Wang, Amin Saied, Weizhu Chen, and Nan Duan. 2023. AGIEval: A Human-Centric Benchmark for Evaluating Foundation Models. arXiv:2304.06364 [cs.CL] https://arxiv.org/abs/2304.06364

[236] Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, and Yu-Xiong Wang. 2024. Language Agent Tree Search Unifies Reasoning Acting and Planning in Language Models. arXiv:2310.04406 [cs.AI] https://arxiv.org/abs/2310.04406

[237] Chunting Zhou, Pengfei Liu, Puxin Xu, Srini Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, Susan Zhang, Gargi Ghosh, Mike Lewis, Luke Zettlemoyer, and Omer Levy. 2023. LIMA: Less Is More for Alignment. doi:10.48550/arXiv.2305.11206 arXiv:2305.11206 [cs].

[238] Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc V Le, and Ed H. Chi. 2023. Least-to-Most Prompting Enables Complex Reasoning in Large Language Models. In *The Eleventh International Conference on Learning Representations*. https://openreview.net/forum?id=WZH7099tgfM

[239] Heng Zhou, Hejia Geng, Xiangyuan Xue, Zhenfei Yin, and Lei Bai. 2025. ReSo: A Reward-driven Self-organizing LLM-based Multi-Agent System for Reasoning Tasks. *arXiv preprint arXiv:2503.02390* (2025).

[240] Xueyang Zhou, Guiyao Tie, Guowen Zhang, Weidong Wang, Zhigang Zuo, Di Wu, Duanfeng Chu, Pan Zhou, Lichao Sun, and Neil Zhenqiang Gong. 2025. Large Reasoning Models in Agent Scenarios: Exploring the Necessity of Reasoning Capabilities. *arXiv preprint arXiv:2503.11074* (2025).

[241] Zihao Zhou, Shudong Liu, Maizhen Ning, Wei Liu, Jindong Wang, Derek F. Wong, Xiaowei Huang, Qiufeng Wang, and Kaizhu Huang. 2025. Is Your Model Really A Good Math Reasoner? Evaluating Mathematical Reasoning with Checklist. In *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=nDvgHIBRxQ

[242] Zixuan Zhou, Xuefei Ning, Ke Hong, Tianyu Fu, Jiaming Xu, Shiyao Li, Yuming Lou, Luning Wang, Zhihang Yuan, Xiuhong Li, Shengen Yan, Guohao Dai, Xiao-Ping Zhang, Yuhan Dong, and Yu Wang. 2024. A Survey on Efficient Inference for Large Language Models. arXiv:2404.14294 [cs.CL] https://arxiv.org/abs/2404.14294

[243] Jiace Zhu, Yingtao Shen, Jie Zhao, and An Zou. 2025. Path-Consistency: Prefix Enhancement for Efficient Inference in LLM. doi:10.48550/arXiv.2409.01281 arXiv:2409.01281 [cs].

[244] Yichen Zhu, Minjie Zhu, Ning Liu, Zhiyuan Xu, and Yaxin Peng. 2024. Llava-phi: Efficient multi-modal assistant with small language model. In *Proceedings of the 1st International Workshop on Efficient Multimedia Computing under Limited*. 18–22.